

Literature Mining Report

Query Statistics

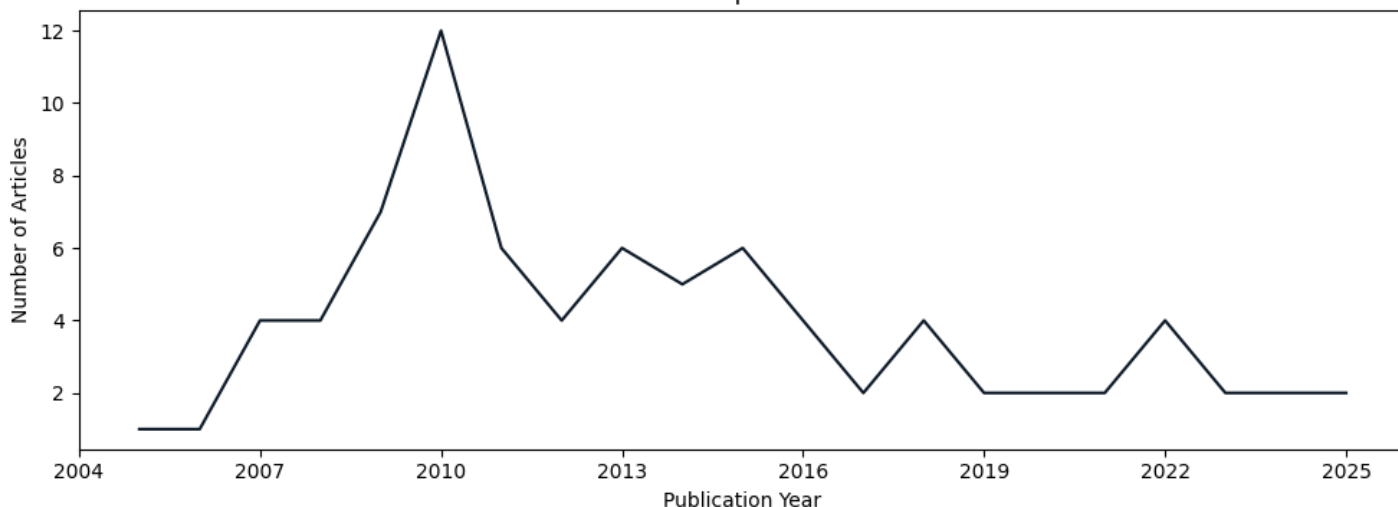
Query date: 2025-08-14

Query terms: "ASE asymmetry c elegans lsy-6"

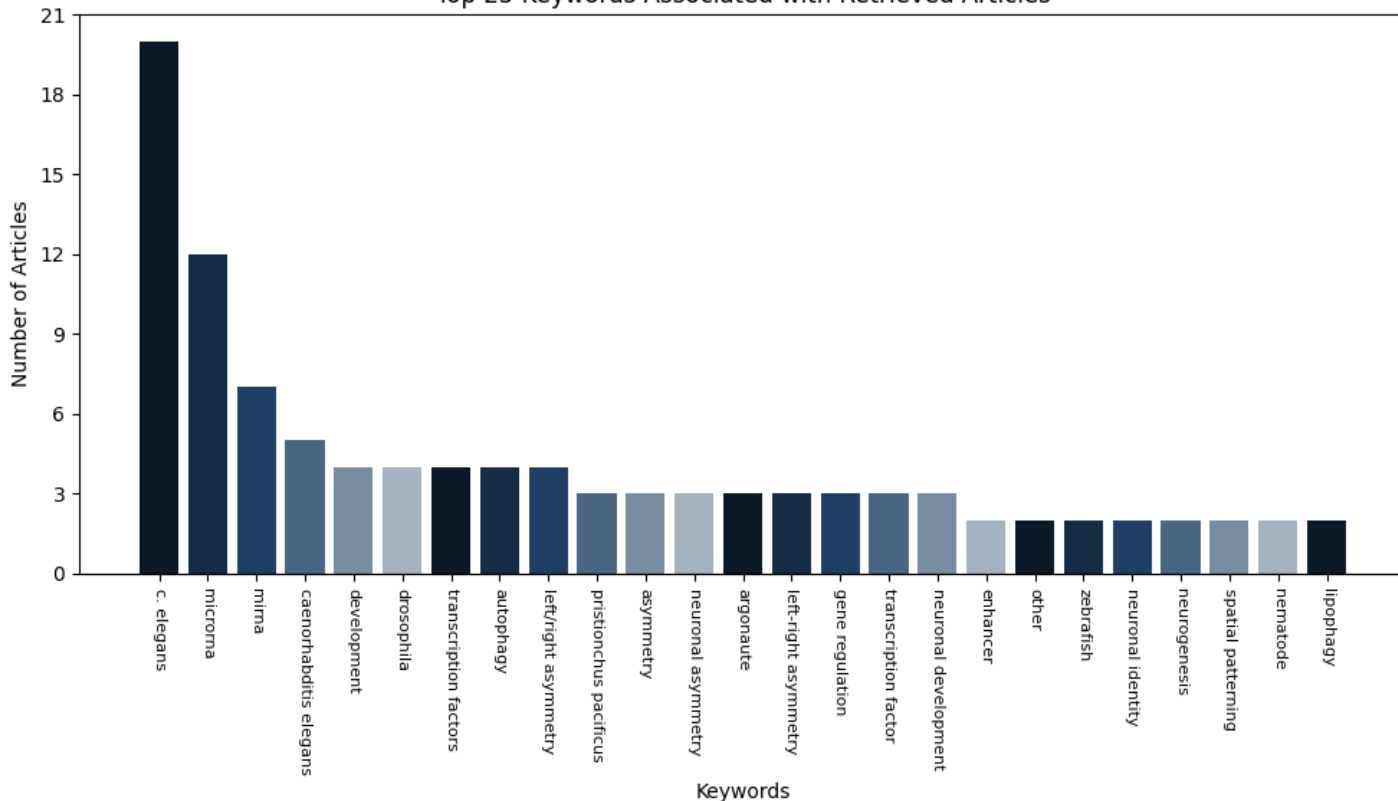
Articles parsed/hits: 82/82 (100.00%)

Scoring keywords: ASE, che-1, lsy-6

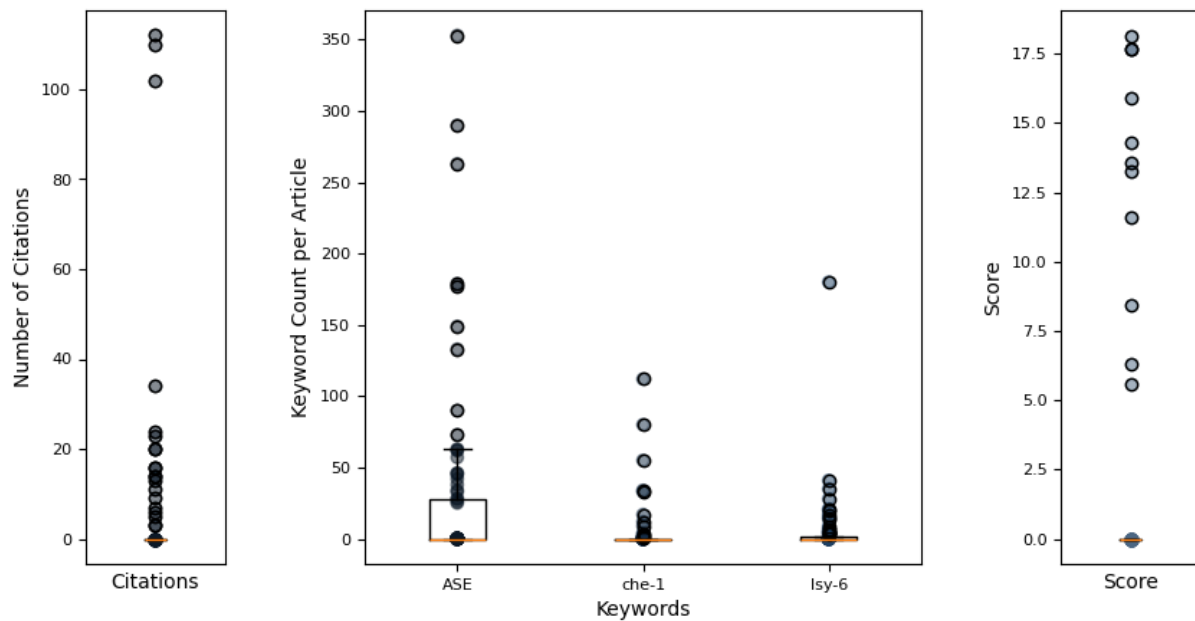
Articles Retrieved per Publication Year



Top 25 Keywords Associated with Retrieved Articles



Hermes Report Summary



Query Results

PMID	Year	Journal	Authors	Title	Citations	Score
33002421	2020	Developmental C	Charest, Julien et al.	combinatorial action of temporally segregated...	0	18.15
40586702	2025	eLife	NA	evolution of lateralized gustation in nematod...	3	17.65
39167071	2024	Genetics	Poole, Richard et al.	Neurogenesis in Caenorhabditis elegans	7	17.64
31526477	2019	eLife	Hong, Ray L et al.	evolution of neuronal anatomy and circuitry i...	34	15.91
39282255	2025	bioRxiv	Mackie, Marisa et al.	evolution of lateralized gustation in nematod...	0	14.27
21698137	2011	PLoS Genetics	Poole, Richard et al.	a genome-wide rnai screen for factors involve...	0	13.58
34504291	2021	Communications	Dekkers, Martij et al.	plasticity in gustatory and nociceptive neuro...	9	13.24
19875417	2010	Nucleic Acids R	Takayama, Jun et al.	single-cell transcriptional analysis of taste...	0	11.62
24065887	2013	Frontiers in Ce	Alqadah, Amel et al.	microrna function in left-right neuronal asym...	0	8.43
29481550	2018	Nature methods	Alberti, Chiara et al.	cell-type specific sequencing of micrnas fr...	0	6.28
26635721	2015	Frontiers in Ne	Davis, Gregory et al.	micrnas: not fine-tuners but key regulato...	0	5.57
20303342	2010	Brain research	Miller, Brooke et al.	MicroRNA dysregulation in psychiatric disease	0	0.00
21737278	2011	Current biology	Bertrand, Vince et al.	notch-dependent induction of left/right asymm...	0	0.00
37438210	2024	Seminars in cel	Destain, Honori et al.	maintenance of neuronal identity in c. elegan...	0	0.00
29967059	2018	Genetics	Ambros, Victor et al.	recent molecular genetic explorations of caen...	0	0.00
29709599	2018	Developmental b	Palmisano, Nich et al.	Autophagy in C. elegans development	24	0.00
35392413	2022	Frontiers in Ne	Chen, Yen-Chung et al.	integration of spatial and temporal patternin...	0	0.00
21338882	2011	Neuron	Chen, Jun-An et al.	mir-17-3p controls spinal neural progenitor p...	0	0.00
21624898	2011	Briefings in Fu	Jo, Jeanyoung et al.	a growing molecular toolbox for the functiona...	0	0.00
18388284	2008	Science (New Yo	Losick, Richard et al.	Stochasticity and Cell Fate	0	0.00
26050091	2015	Protein Cell	Lin, Long et al.	lsy-2 is essential for maintaining the germ-s...	0	0.00
19259264	2009	PLoS ONE	Tursun, Baris et al.	a toolkit and robust pipeline for the generat...	0	0.00
22118276	2011	Methods in cell	Wang, Zheng et al.	Dissection of Genetic Pathways in C. elegans	0	0.00
19060335	2009	Development (Ca	Etchberger, Joh et al.	cis-regulatory mechanisms of left/right asymm...	0	0.00
27095199	2016	Nucleic Acids R	Flamand, Mathie et al.	poly(a)-binding proteins are required for mic...	0	0.00

Query Result 1/25**Combinatorial Action of Temporally Segregated Transcription Factors**

Charest, Julien; Daniele, Thomas; Wang, Jingkui; Bykov, Aleksandr; Mandlbauer, Ariane; Asparuhova, Mila; Rhsner, Josef; Gutierrez-Prez, Paula; Cochella, Luisa

Developmental Cell

Year: 2020 PMID: 33002421 doi: 10.1016/j.devcel.2020.09.002 Citations: 0 Score: 18.15

Keywords hits: ASE: 179, che-1: 55, Isy-6: 18

Abstract:

Combinatorial action of transcription factors (TFs) with partially overlapping expression is a widespread strategy to generate novel gene-expression patterns and, thus, cellular diversity. Known mechanisms underlying combinatorial activity require co-expression of TFs within the same cell. Here, we describe the mechanism by which two TFs that are never co-expressed generate a new, intersectional expression pattern in *C.elegans* embryos: lineage-specific priming of a gene by a transiently expressed TF generates a unique intersection with a second TF acting on the same gene four cell divisions later; the second TF is expressed in multiple cells but only activates transcription in those where priming occurred. Early induction of active transcription is necessary and sufficient to establish a competent state, maintained by broadly expressed regulators in the absence of the initial trigger. We uncover additional cells diversified through this mechanism. Our findings define a mechanism for combinatorial TF activity with important implications for generation of cell-type diversity.

Summary:

Thus, ASEL-specific expression of *Isy-6* cannot be determined solely by *CHE-1*: it depends instead on an intersectional strategy defined by the different lineage histories of the two ASE neurons, which diverge at the 4-cell stage of embryogenesis, eight cell divisions prior to the birth of the ASEs. The two lineage branches that give rise to the ASE neurons are distinguished by exclusive expression of two redundant T-box TFs, *TBX-37/38*, in the ASEL precursor lineage, *ABa*. Focusing on the *Isy-6* locus, we observed that both *TBX-37/38* bound downstream of *Isy-6*, overlapping with the cis-regulatory element that had been genetically defined. Moreover, although *TBX-37* was expressed throughout the embryo, YFP was expressed exclusively in the ASE neurons, supporting that *Isy-6* specificity is generated by the intersection with the ASE-specific *CHE-1*. Together, these data show that early, direct action of *TBX-37/38* on the *Isy-6* locus is necessary and sufficient to prime *Isy-6* for subsequent activation by *CHE-1* in the ASEs. *TBX-37/38* Are Transiently Required to Prime *Isy-6* Previous work suggested that *TBX-37/38* expression is temporally separated from that of *CHE-1*, but it has remained unclear whether low levels of *TBX-37/38* persist and continue to act on *Isy-6* at later time points. We conclude that early adoption of a transcriptionally active state is necessary and sufficient to prime *Isy-6* for later robust activation and explains the requirement for *TBX-37/38*. The Activity of *TBX-37/38* Is Not Relayed by Other T-Box Transcription Factors Given that *TBX-37/38* bind the *Isy-6* locus early and are not continuously required, we set out to investigate how this competent state of *Isy-6* is maintained. Broadly acting TFs could promote low levels of transcription of the *Isy-6* locus that, together with the active chromatin marks, may propagate the competent state until the onset of *CHE-1*. Temporal Integration of Transcriptional Inputs as a General Mechanism for Cell Diversification Priming of enhancers prior to the time point when robust transcription is required is a broadly observed phenomenon that can be caused by different mechanisms, e. g., pre-establishment of contact between promoter and distal enhancers, action of pioneer TFs, acquisition of chromatin modifications, or as shown here by promoting early antisense transcription.

Mentioned biomedical entities:

Genes: *ASER*, *C32C4.16*, *CHE-1*, *Cas9*, *FoxA*, *L1*, *MYST* histone acetyltransferase, *PHA-4*, *TBX-37*, *TBX-38*, *TEL*, *TF1*, *ZIF-1*, *Zdbf2*, *galK*, *gcy-5*, histone, *Isy-6*, *unc-119*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 28-cell, *ABa*, *ABa* lineage, *ABa*-derived cells, *ABp*, *ABp* lineage, *ABp*-derived cells, *ABp*-derived lineage, ASEL neuron, AVE, AssaysAMPure XP BeadsBeckman, Blastomere, C, Cell, D lineages, *E. coli*, EC, Egg ExtractionMLC148, Eggs, Fosmid clones, HT115

StrainCGCHT115Escherichia coli, Hsu, Lineage, Lineage Analysis2-cell embryos, Neuron, Oil, Plasmid, SW15 E., T-box TFs, T4, *TBX-37/38*, TF, TF *CHE-1*, TF GAL4, TF *Isy-2*, TFs, YFP, cell, cell-specific, cell-type, cells, cellular, clones, coli HT115 clones, cytosol, embryonic cells, embryos, epidermal cells, extrachromosomal lines, extraembryonic lineages, fluorescence-activated cell, larvae, leaves cells, left, line, lineage, lineage-specific, lineage-specific TFs, *Isy-2*, neuroblast, neuron, neuronal, neurons, post-mitotic ASEL, progenitor, progenitor cells, sensory-neuron, single-cell, zygote

Organisms: 2-cells embryos, *C. E.*, HLH-14, Noma, Sola SM2 solid, *elegans* lineage, gonad, mouse, mouse embryos, pharynx

Tissues: *ABp* lineage, ASEL precursor lineage, *TBX-37/38*, capillary mouth, egg, embryo, embryos, gut endoderm, mesodermal, microcentrifuge tube, microfuge tube, neural crest

Pathways: N/A

Query Result 2/25

Evolution of lateralized gustation in nematodes

NA

eLife

Year: 2025

PMID: 40586702

doi: 10.7554/eLife.103796.3.sa0

Citations: 3

Score: 17.65

Keywords hits: ASE: 177, che-1: 112, Isy-6: 7

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in *Pristionchus pacificus* respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode *Caenorhabditis elegans*. Previously we showed that *P. pacificus* likely lacked bilateral asymmetry (Hong et al., 2019). Here, we show that by visualizing neuronal activity patterns, contrary to previous expectations based on its genome sequence, the salt responses of *P. pacificus* are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Pacificus che-1 expression in ASE and AFD amphid neurons. The *che-1::GFP* marker in the *che-1::RCaMP* reporter strain is expressed in the ASE and AFD amphid neurons. Since NH_4I sensation is affected by silencing of *che-1* neurons but is unaffected in *che-1* mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of *che-1*. The ASE neurons show left/right asymmetric responses to salt. To assess whether *Ppa* ASE neurons show the same lateralized response to salt ions as *Cel* ASE neurons, we generated transgenic *P. pacificus* lines that express *RCaMP* in ASE neurons and assessed calcium responses to attractive salts. Salts were presented at 1 s for a duration of 2 s, and then removed for the remaining 3 s the total recording time was 6 s. *ASEL* and *ASER* neuron responses to high compared to low concentrations of NH_4Cl . Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neurons response toward 25 mM NH_4Cl , NaCl, and NH_4I , the AFD responses were more robust than *ASER* toward 25 mM NH_4Cl , NaCl, and NH_4I .

Mentioned biomedical entities:

Genes: ADF, ASER, B, CHE-1, Cas9, Cel, GCY receptor guanylyl cyclase, HisCl1, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, foz-1, gcy-5, lim-6, lin-4, Isy-6, rGC, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AFD, AFD amphid neurons, ASE, ASEL-type, B) ASEL, C, E. coli, F, F) 25, F) ASEL, F) NH_4I , F, I, F;, Lo, Mori, NIH, NIH SC1GM1497, NRK, RFP+, RFP-, amphid neuron cell, cell lineage, cells, cellular, dorsal, embryonic Notch, gland cell, left, nervous, neural, neurite, neuron, neuronal, neurons, pacificus, pacificus lines, single-cell, vertical lines

Organisms: *C. elegans*, *C. elegans* ASE taste neurons, *C. elegans* ASER, *C. elegans* ASER neuron, *C. elegans* CHE-1, *C. elegans che-1* mutant, *C. elegans gcy-22*, *C. elegans* neurons, Kato, human, nose

Tissues: AWB neurons, amphid neurons, amphid sensory neurons the first, dorsal, dorsal lateral midline contralaterally, inner channel, lateral, nervous, pharyngeal marginal cells, sensory palates

Pathways: N/A

Query Result 3/25**Neurogenesis in *Caenorhabditis elegans***

Poole, Richard J; Flames, Nuria; Cochella, Luisa; Sundaram, M

Genetics

Year: 2024

PMID: 39167071

doi: 10.1093/genetics/iyae116

Citations: 7

Score: 17.64

Keywords hits: ASE: 149, che-1: 34, lsy-6: 17

Abstract:

Animals rely on their nervous systems to process sensory inputs, integrate these with internal signals, and produce behavioral outputs. This is enabled by the highly specialized morphologies and functions of neurons. Neuronal cells share multiple structural and physiological features, but they also come in a large diversity of types or classes that give the nervous system its broad range of functions and plasticity. This diversity, first recognized over a century ago, spurred classification efforts based on morphology, function, and molecular criteria. *Caenorhabditis elegans*, with its precisely mapped nervous system at the anatomical level, an extensive molecular description of most of its neurons, and its genetic amenability, has been a prime model for understanding how neurons develop and diversify at a mechanistic level. Here, we review the gene regulatory mechanisms driving neurogenesis and the diversification of neuron classes and subclasses in *C. elegans*. We discuss our current understanding of the specification of neuronal progenitors and their differentiation in terms of the transcription factors involved and ensuing changes in gene expression and chromatin landscape. The central theme that has emerged is that the identity of a neuron is defined by modules of gene batteries that are under control of parallel yet interconnected regulatory mechanisms. We focus on how, to achieve these terminal identities, cells integrate information along their developmental lineages. Moreover, we discuss how neurons are diversified postembryonically in a time-, genetic sex-, and activity-dependent manner. Finally, we discuss how the understanding of neuronal development can provide insights into the evolution of neuronal diversity.

Summary:

And how do the differences in the history of 2 convergent cells impact their terminal identity? Below, we discuss some of the regulatory mechanisms that lead to the expression of specific combinations of terminal selectors and other regulators in specific branches of the lineage. The Wnt asymmetry pathway is a general mechanism for diversifying cells along the lineage. The Wnt/catenin asymmetry pathway is a central mechanism for defining changes in gene expression associated with most asymmetric cell divisions during *C. elegans* development (Fig. 1). Importantly, conserved members of the heterochronic pathway, such as *lin-28* and *let-7* and the more recently characterized regulator of *LIN-28*, *makorin/lep-2*, have been found to be involved in timing of sexual maturation in mammals. Some examples of sexual maturation in the *C. elegans* nervous system are as follows: the maturation of the hermaphrodite-specific HSNs and the male-specific CEM neurons at the L4 stage, the downregulation of the olfactory receptor *odr-1* in adult male AWA neurons to promote exploration over feeding and the induction of the TGF- β ligand *daf-7* in adult male ASJ neurons that promotes mate-searching behavior. In *C. elegans*, there are also 2 phases of HOX cluster gene importance: in the early specification of neuronal precursors and the AP patterning of postembryonic neuronal lineages, but also later during the acquisition of specific neuronal features. With regard to the functions in neuroblast specification, single and double mutant analyses have indicated that HOX cluster genes influence both the timing and number of cell divisions of several embryonic neuroblasts in a manner analogous to observations in *Drosophila* and vertebrate neural stem cells. As discussed below, many but not all these genes have been shown to regulate neural vs nonneural decisions and the specification of neural progenitors. The discovery in *Drosophila* that *achaetescute* controls the development of the external sensory organ precursors and that *atonal* controls the development of the internal chordotonal organs led to the realization that proneural genes not only regulate the specification of neuronal precursors but may also play important roles in the differentiation of different classes of neurons. This classic proneural function is not restricted to postembryonic lineages as *lin-32* is also expressed and required in the ABplaaaa/ABarpapaa neuroblasts that give rise respectively to the left and right URAD, CEPD, and URX neurons. Several other *C. elegans* proneural genes have also been shown to regulate the specification of neuroblasts including the AS-C homolog *hlh-14*, which is required for the embryonic specification of the PVQ/HSN/PHB neuroblasts, the ABalpppp/ABpraaap neuroblasts, the DVC neuroblast, and the PVR neuron. The Neuro D homolog *cnd-1*, which is required for the specification of the neuroblasts that give rise to the embryonic DA, DB, and DD ventral nerve cord motor neurons and the Neurogenin homolog *ngn-1*, which is required for the LR asymmetric specification of the MI neuron.

Mentioned biomedical entities:

Genes: ADF, ASE, ASER, AST-1, Arf, B, BAG, CDK8, CEH-1, CEH-13, CEH-14, CEH-34, CEH-36, CHE-1, COG-1, CRH-1, CUT, CaMKII, CtBP, DAF-16, DAF-19, DAF-2, DDs, DIE-1, DMD-3, EGL-44, EGL-5, FACT, FLP, GLR, HD, HES, HLH-2, HLH-3, HLH-6, Hh, Histone, IL2, L1, L2, L4, LAG-1, LIM, LIN-11, LIN-14, LIN-26, LIN-28, LIN-29, LIN-32, LIN-39, LIN-41, Lin28, LinA, MAB-3, MEC-3, MEF-2, MEF2, MELK, MKRN3, Myt1, MyoD, Myt1, NGF, NHR, Nanog, Neurogenin, Notch, ONECUT, Olig, Otx, P4, P4, PA, PAG-3, PHB, POP-1, REF-1, REF-2, RFX, RFX transcription factor, RIP, SOX, SOX-2, SYS-1, Sox, Sox-3, Sox2, SoxB, SoxC, SunS, TBX-37, TBX-38, TCF, TRA-1, TTX-3, TanK, UNC-13, UNC-18, UNC-3, UNC-3, UNC-37, UNC-39, UNC-4, UNC-42, UNC-55, UNC-62, UNC-86, VAB-7, VD, VGLUT, VNC, Wnt, ZAG-1, ast-1, cAMP, cah-3, ced-3 caspase, ceh-1, ceh-12, ceh-13, ceh-17, ceh-2, ceh-2, ceh-23, ceh-24, ceh-28, ceh-3, ceh-32, ceh-34, ceh-37, ceh-7, ces-1, ces-2, cfi-1, cog-1, die-1, egl-13,

egl-18, egl-44, egl-46, egl-5, elt-1, eor-1, histone, hlh-16, hlh-17, hlh-3, hlh-6, let-7, lim-4, lim-6, lin-11, lin-14, lin-26, lin-28, lin-32, lin-39, lin-4, lsy-6, mab-5, mab-9, mec-3, ngn-1, nhr-67, nob-1, odr-1, php-3, pros-1, sem-4, sox, sox-2, str-2, str-44 receptor, ttx-1, unc-13, unc-3, unc-3, unc-39, unc-4, vab-15, vab-7

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 4-cell, 45 cell, A cellular, AB lineage, ABa, ABa blastomere, ADL lineage, AFD, ALM neuron, AMsh glial cells, AMso cells, AMso glial cells, ASE, AWA neurons, BMC Evol Biol, Bayer, C-lineage, CEM, CEPsh glia, CNS stem cells, CUT TFs, Cell, Cell Dev Biol, Cell Rep., Cell Tissue Res, Cells, Cellular, D lineage, DAs, E lineage, FKH-8 TFs, GLR cell, GLR glia, Glia, Glial, HBL-1, HD TF, HD TFs, HSN, HSN lineage, Hobert 218).ASJ neurons, ILso glial cells, Int J Dev Biol., K blast cell, Kenyon, L/R lineage, L1, L2, L2 Caenorhabditis, L4, L4 hermaphrodite neurons, L4 neurons, NIH, NRH TF, NSM, Neural, Neuron, Neuronal, Neuronal cell lineages, Neurons, P11 cells, PHso cells, Post-embryonic cell, Postembryonic neural progenitors, Q lineage, Q neuroblasts, Sasai, Sci Rep. 7(1):7294, Single-cell, Stem Cells, T seam cells, TBX-37/38, TF, TF UNC-4, TF unc-3, TFs, TRN, URX neurons, V5 neuroblasts, adult C. elegans motor neurons, adult neural stem cells, anterior cell, anterior seam cells, astrocytes, astroglia, b) L4, b) Lineage, bipotent neuromesodermal progenitors, blast cell, blast cells, blastomere, blastomeres, cardiac cell, ceh-43, ceh-44 CUX HD, cell, cell lineage, cell lineage-specific, cell-specific, cell-type, cells, cellular, chemoreceptor, cilia, cilium, class-specific, colspan=1/AWA, couples cell lineage, e3D epithelial cell, ectoderm, ectodermal, ectodermal blast cells, ectodermal cells, ectodermal germ layer, embryonic blastomeres, embryonic cell lineage, embryonic lineage, embryonic neural, embryonic neuroblast cell, embryonic neuroblasts, embryonic neuronal, endodermal, endodermal gut cells, ensheath, epidermal cell progenitor, epidermal cells, epithelial cells, fly, germ cell, germ layers, gland cells, glia, glia-to-neuron, glial, glial cell, glial cells, hermaphrodite-specific, hermaphrodites, hypodermal, hypodermal cell, hypodermal cells, intestinal cells, larvae, left, lineage, lineage-specific TFs, lines, male-specific glia, mesoderm, mesodermal, mesodermal lineage, migrating cells, motoneurons, muscle cell, muscle cells, nervous, neural, neural cell, neural cells, neural progenitor, neural progenitor cell, neural progenitors, neural stem, neural stem cell, neural stem cell lineages, neural stem cells, neural/mesodermal cell, neurite, neuroblast, neuroblast lineage, neuroblasts, neuroendocrine, neuron, neuron class-specific, neuron class-specific TFs, neuron class-specific terminal selectors, neuron-type, neuronal, neuronal cell, neuronal cell type, neuronal cell-type, neuronal cells, neuronal ectoderm, neuronal lineage, neuronal progenitor, neuronal progenitors, neurons, non-neuronal, non-neuronal ectodermal cell, nonneural, nonneural cell, nonneural ectoderm, nonneural ectodermal, nonneural ectodermal cell, nonneural ectodermal cells, nonneural ectodermal derivatives, nonneuronal cell, nonneuronal cells, nonneuronal ectodermal fates, nonneuronal progenitors, nonneuronal tissues, parasite-specific, postembryonic blast cells, posterior cell, posterior cells, posterior daughter cells, postmitotic cells, postmitotic neural, postmitotic neuronal, progenitor cells, progenitors, proneural TFs, rectal epithelial cell, rectal epithelial cell K, rectal epithelial cells, rectal epithelial cells K, seam cells, secretory cells, sensory-neuron, single-cell, socket glia, somatic cell, somatic cells, synapses, terminal TFs, type-D GABAergic neuron, ventral epidermal cells, zag-1, ztf-11, zygote, 2/cell

Organisms: ADL chemosensory neurons, ALN sensory neurons, Barr, C. elegans, C. elegans UNC-3 homeodomain protein, C. elegans ceh-36, C. elegans contains 2 SoxB, C. elegans embryonic, C. elegans genome, C. elegans germ cells, C. elegans glia, C. elegans homeobox, C. elegans insulin, C. elegans lineage, C. elegans mesodermal glia, C. elegans nervous system, C. elegans neuronal, C. elegans neurons, C. elegans proneural bHLH TFs, Caenorhabditis elegans nervous system, Caenorhabditis elegans neuronal, Hart, Mouse, V5 blast cell, body, brain, chick, embryonic central nervous system, enteric nervous, human, human basic helix-loop-helix proteins, human cells, humans, intestine, male-specific nervous system, mice, mouse, mouse brain, mouse nervous system, nervous, organ, organs, peripheral nervous, pharynx, skin, unc-47 GABA vesicular transporter, ventral midline, zebrafish

Tissues: ABalppp/praaa neural precursor, AFD thermosensory neuron, AIY interneuron, AIY interneuron pair, ASJ polymodal neurons, AWC olfactory neurons, AWC sensory neuron type, AWC sensory neurons, CAN neurons, PDA motor neuron, Parasite-specific neural, VC5 hermaphrodite-specific, amphid sensory organs, anterior, body, cuticle, dorsal, dorsal muscles by V-type, egg, endings, endothelial, epithelial, hermaphrodite-specific HSNs, hermaphrodite-specific motor neurons, hypodermal, hypodermis, innervate dorsal muscles, lateral, mammalian spinal cord motor neuron, mechanosensory channel, mechanosensory neurons, mesodermal body wall muscles, muscle, muscles, nervous, nervous system, nervous system arise, nervous system based on electron microscopy (, nervous system depend heavily, nervous system gene regulatory networks., nervous system whose, neural, neural ectodermal, neural precursor, neural tissue, nonneuronal tissue, olfactory, peripheral, pharyngeal muscle cell, post-mitotic nervous system, rectal epithelial blast cells, secretory pharyngeal gland cells, serotonergic chemosensory neurons, sex-shared AWA olfactory neurons, sex-shared neurons, striated muscle, thermosensory neuron, tissue, tissues

Pathways: N/A

Query Result 4/25**Evolution of neuronal anatomy and circuitry in two highly divergent nematode species**

Hong, Ray L; Riebesell, Metta; Bumbarger, Daniel J; Cook, Steven J; Carstensen, Heather R; Sarpolaki, Tahmineh; Cochella, Luisa; Castrejon, Jessica; Moreno, Eduardo; Sieriebriennikov, Bogdan; Hobert, Oliver; Sommer, Ralf J; Sengupta, Piali; Marder, Eve; Sengupta, Piali

eLife

Year: 2019 PMID: 31526477 doi: 10.7554/eLife.47155.053 Citations: 34 Score: 15.91

Keywords hits: ASE: 9, che-1: 33, lsy-6: 4

Abstract:

The nematodes *C. elegans* and *P. pacificus* populate diverse habitats and display distinct patterns of behavior. To understand how their nervous systems have diverged, we undertook a detailed examination of the neuroanatomy of the chemosensory system of *P. pacificus*. Using independent features such as cell body position, axon projections and lipophilic dye uptake, we have assigned homologies between the amphid neurons, their first-layer interneurons, and several internal receptor neurons of *P. pacificus* and *C. elegans*. We found that neuronal number and soma position are highly conserved. However, the morphological elaborations of several amphid cilia are different between them, most notably in the absence of winged cilia morphology in *P. pacificus*. We established a synaptic wiring diagram of amphid sensory neurons and amphid interneurons in *P. pacificus* and found striking patterns of conservation and divergence in connectivity relative to *C. elegans*, but very little changes in relative neighborhood of neuronal processes. These findings demonstrate the existence of several constraints in patterning the nervous system and suggest that major substrates for evolutionary novelty lie in the alterations of dendritic structures and synaptic connectivity.

Summary:

All amphid neurons and the amphid sheath cell in *P. pacificus* have their cell bodies in the lateral ganglion posterior to the nerve ring, which resembles the condition found in *C. elegans* and other nematodes. Because the wing neurons are the most morphologically distinct and best studied amphid neurons in *C. elegans*, their absence in most other nematodes contributes to the difficulty in assigning homology in the amphid neurons in these other nematode species. 1.7554/eLife.47155.3 Figure 1. EM of the amphid sensillum and two other sensory neurons of *P. pacificus* hermaphrodite adults. All images are from specimen 17, except B and C. As specimens were sectioned from the head, left structures appear on the right side in the images and vice versa. The cell body of the URX neuron in both nematode species is located posterior to the nerve ring, and directly anterior to the anterior-most amphid neuron cell body in dorsal position AM9 in *P. pacificus* and ASK in *C. elegans*. Four of the strong gap junction connections were present in both species while 6 and 2 were specific to *P. pacificus* and *C. elegans*, respectively. Perhaps the main drivers of these chemotaxis differences are changes in olfactory perception, that is sensory anatomy and sensory receptors, rather than the means by which chemosensory signals are processed. A previous comparative analysis of the pharyngeal nervous system, composed of an isolated circuit of 2 neurons, also found that homologous neurons types display differences in synaptic connectivity between *C. elegans* and *P. pacificus*.

Mentioned biomedical entities:

Genes: ADF, AM5, AM6, ASER, B, BAG, CHE-1, FLP, ODR-7 nuclear hormone receptor, PHB, T12, URB, gcy-6, lsy-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AFD, AIA cell bodies, AIY interneuron cell bodies, AMsh, AMsh cell body, AMsh cells, ASE, ASG neurons, AWA cells, AWC cellular, AWx neurons, Amphid Interneuron, Amphid cell bodies, BC, Coates, CD) Dil, E. coli, EMS, F) A Dil, Figure 3videos 115. Figure 3video, HPF, ILso cells, Mori, Neuron, OL, RFP, TEM, TZ cilia, amphid neuron cell, amphid neuronal cilia, amphid sheath cell cytoplasm, amphid sheath cells, amphid socket cells, anterior AMsh endings, anterior cell, anterior-most amphid neuron cell body, bundles, cell, cell body, cell lineage, cell-specific, cell-type, cells, cellular, cilia, ciliated channel neurons, cilium, dorsal-most amphid neurons, dorsal-most cells, elegans-specific, epidermal cell, excretory cell, finger cell, finger cell cilium, finger cells, glia, glia-like amphid sheath cells, glial cells, high-pressure-frozen, lateral, left, line, neuron, neuron-neuron, neuronal, neuronal cell bodies, neurons, pacificus, pacificus FLP neurons, pacificus URX neurons, pacificus cellular homolog, polymodal, posterior AM2 cell, seam cells, secretory cell, sheath cell, sheath cell posterior, sheath cells, sheath glial cells, socket cell, socket cells, soma, somas, synapses, zone, zones

Organisms: AM9, C., *C. elegans*, *C. elegans* ADF, *C. elegans* ADL neuron, *C. elegans* ASE, *C. elegans* BAG, *C. elegans* EM, *C. elegans* genome, *C. elegans* nervous system, *C. elegans* neuronal, *C. elegans* str-2, *C. elegans*-specific receptor protein, *E. coli*, H; Table 3, *Haemonchus contortus*, Hall lab, John White, L1 larvae, Labial, *Parastrongyloides trichosuri* L1, RIB interneurons, Synaptic, labial, nervous, nose, olfactory, organs, pharynx, swine, synaptic, ventral

Tissues: ASJ homologs, FLP neurons, Left lateral view, TEM specimens, URA neurons, URX dendrite, URX endings, URX neuron, URX neurons, amphid, amphid axon, amphid channel, amphid cilia, amphid circuit, amphid dendrites, amphid dendritic endings, amphid finger cell, amphid interneuron, amphid neuron, amphid neuronal, amphid neurons, amphid sensilla, amphid sensilla.(A) Three-dimensional renderings of individual pairs of, amphid sensillum, amphid sensillum., amphid sensory, amphid sensory circuit, amphid sensory circuitry of, amphid sensory circuitry of P. pacificus, amphid sensory neuron, amphid sensory neuron axonal, amphid sensory neuron pairs, amphid sensory neurons, amphid sheath, amphid sheath cell, amphid sheath cell body, amphid sheath cells, amphid sheath channel, amphid sheath gliaTo, amphid sheath processes, amphid sheath, amphid socket, amphid sheath., amphid socket cell, amphid socket cells, amphid wing neurons, anterior, anterior C. elegans, anterior P. pacificus, ciliary elaborations, ciliary endings, cuticle, cuticular sensilla, dendritic endings, dendritic-like endings, dorsal, dorsal amphid neurons, dorsal labial process, dorsal nerve cord, dorsal sheath entry;cell, dorsal tooth (, excretory duct, high-pressure-frozen, inner labial sensilla, inner singlet microtubules, lateral, lateral Inner Labial socket cell, layer, layer amphid, layer amphid interneurons, left sheath cell, left sheath glia, matrix-filled amphid channel, midline, multi-dendritic structures throughout the head region, nervous, nervous system anatomy, nervous system patterning is, non-amphid ciliated, right sheath glia, sensilla, sensory neurons, sheath, sheet-like, socket support cells, specimens, thermosensory neuron ALD, transverse TEM sections, tubule, wing neurons

Pathways: N/A

Query Result 5/25**Evolution of lateralized gustation in nematodes**

Mackie, Marisa; Le, Vivian Vy; Carstensen, Heather R.; Kushnir, Nicole R.; Castro, Dylan L.; Dimov, Ivan M.; Quach, Kathleen T.; Cook, Steven J.; Hobert, Oliver; Chalasani, Sreekanth H.; Hong, Ray L.

bioRxiv

Year: 2025 PMID: 39282255 doi: 10.1101/2024.08.31.610597 Citations: 0 Score: 14.27

Keywords hits: ASE: 133, che-1: 8, Isy-6: 5

Abstract:

Animals with small nervous systems have a limited number of sensory neurons that must encode information from a changing environment. This problem is particularly exacerbated in nematodes that populate a wide variety of distinct ecological niches but only have a few sensory neurons available to encode multiple modalities. How does sensory diversity prevail within this constraint in neuron number? To identify the genetic basis for patterning different nervous systems, we demonstrate that sensory neurons in *Pristionchus pacificus* respond to various salt sensory cues in a manner that is partially distinct from that of the distantly related nematode *Caenorhabditis elegans*. By visualizing neuronal activity patterns, we show that contrary to previous expectations based on its genome sequence, the salt responses of *P. pacificus* are encoded in a left/right asymmetric manner in the bilateral ASE neuron pair. Our study illustrates patterns of evolutionary stability and change in the gustatory system of nematodes.

Summary:

Since NH_4I sensation is affected by silencing of *che-1* neurons but is unaffected in *che-1* mutants, ASE function may be more greatly impacted by the silencing of ASE than by the loss of *che-1*. The ASE neurons show left/right asymmetric responses to salt. To assess whether *Ppa* ASE neurons show the same lateralized response to salt ions as *Cel* ASE neurons, we generated transgenic *P. pacificus* lines that express *RCa MP* in ASE neurons and assessed calcium responses to attractive salts (Fig. Altogether, *P. pacificus* ASE neurons clearly show left/right asymmetric responses to salt attractants and these asymmetric responses show similarities and differences to the *C. elegans* ASE taste neurons. The AFD neurons also respond to salts in *P. pacificus*. The *RCa MP* line that we used to assess calcium responses in ASE is also expressed in AFD, allowing us to simultaneously examine the calcium responses in the AFD neurons (Fig. Specifically, although we observed weaker or comparable responses in AFD neurons when compared to either ASE neurons response towards 25 mM NH_4Cl , NaCl and NH_4I (Fig. However, the responses to 25 mM NaCl were not significantly reduced in either the *ASEL* or *ASER* neuron in the *gcy-22.3* mutant (Fig. Nevertheless, we find that *P. pacificus* neurons have a distinct response to this salt when compared to those observed in *C. elegans* neurons.

Mentioned biomedical entities:

Genes: ADF, ASER, CHE-1, Cas9, Cel, GCY receptor guanylyl cyclase, HisCl1, OTX, Ppa, RIP, SHC, TTX-1, Twist, cog-1, die-1, foz-1, gcy-5, lim-6, lin-4, Isy-6, rGC, ttx-1

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AFD, ASE, ASEL-type, *E. coli*, NIH, NIH SC1GM1497, NRK, Nematode Strains *P. pacificus*, cell lineage, cells, cellular, embryonic Notch, gland cell, left, nervous, neural, neurite, neuron, neuronal, neurons, newly-generated, *pacificus*, *pacificus* lines, single-cell

Organisms: *C. elegans*, *C. elegans* ASE taste neurons, *C. elegans* ASER, *C. elegans* ASER neuron, *C. elegans* CHE-1, *C. elegans* *che-1* mutant, *C. elegans* *gcy-22*, *C. elegans* neurons, human, nose

Tissues: AWB neurons, Sartorius, amphid neurons, amphid neurons (, amphid sensory neurons the first, lateral, nervous, pharyngeal marginal cells, sensory palates

Pathways: N/A

Query Result 6/25

A Genome-Wide RNAi Screen for Factors Involved in Neuronal Specification in *Caenorhabditis elegans*, bScreen for Neuronal Fate Mutants

Poole, Richard J.; Bashllari, Enkelejda; Cochella, Luisa; Flowers, Eileen B.; Hobert, Oliver; Mango, Susan E.

PLoS Genetics

Year: 2011 PMID: 21698137 doi: 10.1371/journal.pgen.1002109 Citations: 0 Score: 13.58

Keywords hits: ASE: 353, che-1: 12, Isy-6: 35

Abstract:

One of the central goals of developmental neurobiology is to describe and understand the multi-tiered molecular events that control the progression of a fertilized egg to a terminally differentiated neuron. In the nematode *Caenorhabditis elegans*, the progression from egg to terminally differentiated neuron has been visually traced by lineage analysis. For example, the two gustatory neurons ASEL and ASER, a bilaterally symmetric neuron pair that is functionally lateralized, are generated from a fertilized egg through an invariant sequence of 11 cellular cleavages that occur stereotypically along specific cleavage planes. Molecular events that occur along this developmental pathway are only superficially understood. We take here an unbiased, genome-wide approach to identify genes that may act at any stage to ensure the correct differentiation of ASEL. Screening a genome-wide RNAi library that knocks-down 18,179 genes (94% of the genome), we identified 245 genes that affect the development of the ASEL neuron, such that the neuron is either not generated, its fate is converted to that of another cell, or cells from other lineage branches now adopt ASEL fate. We analyze in detail two factors that we identify from this screen: (1) the proneural gene *hlh-14*, which we find to be bilaterally expressed in the ASEL/R lineages despite their asymmetric lineage origins and which we find is required to generate neurons from several lineage branches including the ASE neurons, and (2) the COMPASS histone methyltransferase complex, which we find to be a critical embryonic inducer of ASEL/R asymmetry, acting upstream of the previously identified miRNA *Isy-6*. Our study represents the first comprehensive, genome-wide analysis of a single neuronal cell fate decision. The results of this analysis provide a starting point for future studies that will eventually lead to a more complete understanding of how individual neuronal cell types are generated from a single-cell embryo.

Summary:

While both *mex-3* and *mex-5* led to an almost complete loss of neurons, no other gene examined generated such a complete loss of neurons, suggesting they generate broad neuronal but not pan-neuronal differentiation defects. 1.1371/journal.pgen.1219. t1 Table 1 Genes that elicit broad neuronal defects when knocked-down by RNAi. Gene Name Functional Category Domain/Function ASELNeurons Muscle Gut Hyp ASERASE, *brid*=pgen.1219-Chang1 ref-type=bibr15, 27, and the b HLH gene *hlh-14* that we describe in more detail later in this paper. 1.1371/journal.pgen.1219. t3 Table 3 Genes that elicit a loss of ASE fate when knocked-down by RNAi. Gene Name Functional Category Domain/Function ASELNeurons Muscle Gut Hyp ASERASEViable Mutant phenotype, *brid*=pgen.1219-Flowers1 ref-type=bibr35 by identifying genes with laterality defects in the genetic intervals to which *Isy-15* and *Isy-22* had been mapped, facilitating the identification of the phenotype-inducing mutation within the whole-genome sequencing data. The screen identified an as yet elusive proneural patterning gene for the ASE lineage, the b HLH gene *hlh-14*. Consistent with such a notion we find that on examining ASEL, partial removal of *hlh-14*, through RNAi in animals that express two distinct markers for ASEL fate simultaneously, sometimes leads to loss of one but not the other marker suggesting the ASE cell is still produced but fails to differentiate correctly. Symmetric and asymmetric expression of *hlh-14* in other neuronal lineages In addition to its expression in the ASE lineage, we observe by 4 D-lineage analysis that our *hlh-14* fosmid reporter is expressed in several other neuronal lineages. ASE:: *Isy-6* is *ceh-36prom*:: *Isy-6*. Overlapping expression can be observed. We analyzed several other members of the COMPASS complex that were not revealed by our RNAi screen, using RNAi, mutant alleles or both. Moreover, the 2 ASEL fate inducing activity of *Isy-6*, expressed from a heterologous promoter, is epistatic to the loss of COMPASS activity, corroborating the notion that COMPASS acts upstream of *Isy-6*. Previous genetic screens have revealed that a MYST-type histone acetyltransferase complex also shows ASE laterality defects that are indistinguishable from those of the COMPASS histone methyltransferase complex, in that the ASEL neurons convert into ASER neurons and that *Isy-6* expression is lost in the post-mitotic ASE neurons 33.

Mentioned biomedical entities:

Genes: ASER, Brd, LIN-49, Lon, MYST, NBP, Notch, P, PHD, RbBP5, Unc, WD4, WDR5, WDR82, aph-1, cdk-9, ceh-2, cog-1, cog-1 homeobox, csc-1, csnk-1, die-1, dpy-7, foz-1, histone, klp-16, lim-6, lin-32, lin-49, Isy-6, mag-1, mel-32, mex-3, mog-4, nhr-67, par-3, par-6, pie-1, ref-1, set-2, zyg-9

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 2-cell, 3D cell, ABa, ABalppp/praaa lineage, ABp, ABp lineage, ALM/BDU lineage, ASE, ASE cell, ASE lineage, ASE lineage 44, ASE neuronal,

ASEL fated cells, ASEL lineage, Brenner 1, C, C lineage, C-lineage, C.elegans, Cell, Cells, Clones, Gentra GenePure DNA, I1L/I1R, I1L/I1R blastomeres, I1L/I1R lineage, I1L/I1R neuroblasts, Lineage, NIH, P-cell, P-cells, PVQ/HSN/PHB, PVQ/HSN/PHB lineage, Purple lines, Pvl, RNAi clones, Red cell, TF, TF clone, TFs, V-cells, line 1, anterior-posterior lineage, bacterial clones, basic cellular, blastomere, blastomeres, cell, cells, cellular, clone, clones, ectoderm, embryo neuroblasts, embryonic blastomere AB, end-1/end-3, endoderm, epithelial, extra hyp11 cell, fosmid line, fosmid lines, glia, green cell, gut lineage, his-3, hyp11 cell, hypodermal cell, hypodermal cells, left, lineage, lines, lis-1, mei-2, mesoderm, mesodermal, multi-potent, myo-3prom::mitochondrial, myoepithelial cells, nervous system 1, neural, neural/non-neural, neural/non-neural cell, neuroblast, neuroblast cell, neuroblasts, neuron, neuron cell, neuron-type cell, neuronal, neuronal cell, neuronal cells, neuronal lineage, neurons, neurons/glia, non-neuronal hypodermal cells, pharyngeal muscle cell, purple lines, see, sequence-verified RNAi clones, terminal cell, wells

Organisms: Bristol N2, C. elegans, C. elegans genome, Hobert lab, muscle, nervous, pharynx

Tissues: ABp lineage, ASER neurons, Flowers, anterior, blastomere, body, cuticle, ectodermal, ectodermal tissue, embryo, embryos, hypodermal, hypodermis, meso-, multi-potent pharyngeal, muscle, nervous system, nervous system come, neural, pharyngeal marginal cell (, screenThe tissue, tissue

Pathways: N/A

Query Result 7/25

Plasticity in gustatory and nociceptive neurons controls decision making in *C. elegans* salt navigation

Dekkers, Martijn P. J.; Salfelder, Felix; Sanders, Tom; Umuerri, Oluwatoroti; Cohen, Netta; Jansen, Gert

Communications Biology

Year: 2021

PMID: 34504291

doi: 10.1038/s42003-021-02561-9

Citations: 9

Score: 13.24

Keywords hits: ASE: 29, che-1: 9, Isy-6: 2

Abstract:

A conventional understanding of perception assigns sensory organs the role of capturing the environment. Better sensors result in more accurate encoding of stimuli, allowing for cognitive processing downstream. Here we show that plasticity in sensory neurons mediates a behavioral switch in *C. elegans* between attraction to NaCl in naive animals and avoidance of NaCl in preconditioned animals, called gustatory plasticity. Ca²⁺ imaging in ASE and ASH NaCl sensing neurons reveals multiple cell-autonomous and distributed circuit adaptation mechanisms. A computational model quantitatively accounts for observed behaviors and reveals roles for sensory neurons in the control and modulation of motor behaviors, decision making and navigational strategy. Sensory adaptation dynamically alters the encoding of the environment. Rather than encoding the stimulus directly, therefore, we propose that these *C. elegans* sensors dynamically encode a context-dependent value of the stimulus. Our results demonstrate how adaptive sensory computation can directly control an animals behavioral state.

Summary:

This recovery of the response in ASEL is consistent with behavioral data, where attraction to Na Cl after pre-exposure is restored by a 5-min wash⁶. We conclude that the ASEL neuron desensitizes with pre-exposure to Na Cl and recovers in the absence of Na Cl exposure and suggest that this sensory adaptation modulates the strength of attraction to Na Cl in behavioral assays. Prolonged exposure to Na Cl sensitizes ASH. To test if the response of the ASH neurons is affected by pre-exposure, we first exposed animals to 1 m M Na Cl for 1 min and subsequently introduced a further 1 m M increase to 2 m M Na Cl. D Average Ca²⁺ transients in ASER of eat-4 animals in response to a decrease in Na Cl concentration from 1 m M to m M after 3 or 6 s exposure. Thus, our computational model predicts that the robustness of the performance in the quadrant assay is not reduced by ASE sensitization. In silico ASH sensitization reproduces gustatory plasticity Next, we looked at the behavior of pre-exposed animals focusing on a 15 min 1 m M Na Cl pre-exposure, at which the avoidance behavior is the strongest. 7b), confirming that in our model the separate roles of ASEL and ASER in motor control are direct consequences of the rise and decay times of their responses. Thus, our model results are consistent with results from ASEL and ASER ablated animals, as well as with observations that both steering and pirouettes contribute to the chemotaxis index in the quadrant assay. Accordingly, our model remains agnostic to these possibilities, as the encoding of food in the model is implicit. While multiple sensory neurons and contributions from the downstream circuitry most likely contribute to the rich behavioral responses in *C. elegans*, our simulations demonstrate the feasibility of a parsimonious model in which recruitment of ASH by an ASE derived signal underpins the switch between attraction and avoidance in gustatory plasticity. The bilaterally asymmetric kinetics of the ASE neurons, in which ASEL and ASER depolarize in response to concentration increases and decreases, respectively¹⁷, suggest a potential to double the ASE sensory dynamic range, thus enhancing the resolution of Na Cl sensing in the animal.

Mentioned biomedical entities:

Genes: ADF, ASER, CFP, Isy-6, sra-6

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ASE, CFP, CTX, F, Falcon X, b, b Average, b Average Ca²⁺, b Salt, b, b;, blue lines, cell, cells, cellular, green lines, interneuron, neural, neuron, neuronal, neurons, solid, wild-type, wild-type *C. elegans*

Organisms: ASER Ca²⁺, C., *C. elegans*, gustatory, nervous

Tissues: ASER neurons, amphid sensory neurons, brown, capillary, gustatory, gustatory plasticity², layer, nervous system, plastic edges, taste

Pathways: N/A

Query Result 8/25Single-cell transcriptional analysis of taste sensory neuron pair in *Caenorhabditis elegans*

Takayama, Jun; Faumont, Serge; Kunitomo, Hirofumi; Lockery, Shawn R.; Iino, Yuichi

Nucleic Acids Research

Year: 2010

PMID: 19875417

doi: 10.1093/nar/gkp868

Citations: 0

Score: 11.62

Keywords hits: ASE: 263, che-1: 17, Isy-6: 9

Abstract:

The nervous system is composed of a wide variety of neurons. A description of the transcriptional profiles of each neuron would yield enormous information about the molecular mechanisms that define morphological or functional characteristics. Here we show that RNA isolation from single neurons is feasible by using an optimized mRNA tagging method. This method extracts transcripts in the target cells by co-immunoprecipitation of the complexes of RNA and epitope-tagged poly(A) binding protein expressed specifically in the cells. With this method and genome-wide microarray, we compared the transcriptional profiles of two functionally different neurons in the main *C. elegans* gustatory neuron class ASE. Eight of the 13 known subtype-specific genes were successfully detected. Additionally, we identified nine novel genes including a receptor guanylyl cyclase, secreted proteins, a TRPC channel and uncharacterized genes conserved among nematodes, suggesting the two neurons are substantially different than previously thought. The expression of these novel genes was controlled by the previously known regulatory network for subtype differentiation. We also describe unique motif organization within individual gene groups classified by the expression patterns in ASE. Our study paves the way to the complete catalog of the expression profiles of individual *C. elegans* neurons.

Summary:

The primers used for the amplification of each gene were: *Imn152*: 5-CGTTCCACCACCCACCAGAA-3, *Imn132*:

5-CAAGACGAGCTGATGGGTTATCT-3 for *Imn-1* *gcy551*: 5-CCTACCAAGAGAAAAAGTTGAACTAAGAA-3, *gcy531*:

5-GGGCATGGATAGACCAACGA-3 for *gcy-5* *gcy651*: 5-CACGTGGCGAAGTCATAATCA-3, *gcy6-31*: 5-GCTGACTCGTCCATTTTAAGCA-3 for *gcy-6* *gcy7*. *fw1*: 5-TCTCCCAGACCCGATTTGG-3, *gcy7*. *rv1*: 5-CCGAAGGACGTTTCGGTAAAAC-3 for *gcy-7*. Most of the genes expressed in ASE and/or ASER have ASE motifs in their 5-upstream regions, which are specifically required for the expression in ASE neurons. We found that all of the novel ASER-biased genes had ASE motif in the promoter regions, whereas no ASE motif was found in the ASE-biased *ins-32* promoter. Therefore, we suggest that all of the eight novel ASER-biased genes could be directly regulated by CHE-1 binding to their ASE motifs whereas *ins-32* expression is regulated indirectly. To determine whether the ASE motifs were enriched in the genes expressed in ASE neurons, we examined the frequency of ASE motifs per one kilobase promoter of each gene set categorized by the expression patterns in ASE. The frequency was higher in the genes expressed in either or both of ASE neurons than those not expressed in ASE neurons or those selected randomly from the microarray probe set, suggesting that the ASE motif is enriched in the genes expressed in ASE neurons.

Mentioned biomedical entities:

Genes: ASER, Bad, C14C6.3, CHE-1, L1, PABP, TRP-2 channel, TRPC, cog-1, die-1, flp, foz-1, *gcy-22*, *gcy-5*, *gcy-6*, *ins-32*, *lim-6*, *Isy-6*, *nlp-5*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ASE, BF, BH, Cells, Culture, RFP, cell, cell-specific, cells, chemoreceptor, chemoreceptors, cone photoreceptor cells, embryonic cells, embryonic culture cells, epithelial, fluorescence-activated cell, *lin-44::GFP*, *Imn-1*, muscle cells, neuron, neuronal, neurons, seam cells, single-cell, wild-type

Organisms: *C. elegans*, *C. elegans* ASE neuron class, *C. elegans* lines, *C. elegans* neurons, lino lab, RNA samples, gonad, intestine, muscles

Tissues: ASER neuron, ASER neurons, C7, hypodermis, medial, nervous system, nervous system , tissues

Pathways: N/A

Query Result 9/25microRNA function in left-right neuronal asymmetry: perspectives from *C. elegans*

Alqadah, Amel; Hsieh, Yi-Wen; Chuang, Chiou-Fen

Frontiers in Cellular Neuroscience

Year: 2013

PMID: 24065887

doi: 10.3389/fncel.2013.00158

Citations: 0

Score: 8.43

Keywords hits: ASE: 62, che-1: 3, lsy-6: 2

Abstract:

Left-right asymmetry in anatomical structures and functions of the nervous system is present throughout the animal kingdom. For example, language centers are localized in the left side of the human brain, while spatial recognition functions are found in the right hemisphere in the majority of the population. Disruption of asymmetry in the nervous system is correlated with neurological disorders. Although anatomical and functional asymmetries are observed in mammalian nervous systems, it has been a challenge to identify the molecular basis of these asymmetries. *C. elegans* has emerged as a prime model organism to investigate molecular asymmetries in the nervous system, as it has been shown to display functional asymmetries clearly correlated to asymmetric distribution and regulation of biologically relevant molecules. Small non-coding RNAs have been recently implicated in various aspects of neural development. Here, we review cases in which microRNAs are crucial for establishing left-right asymmetries in the *C. elegans* nervous system. These studies may provide insight into how molecular and functional asymmetries are established in the human brain.

Summary:

In addition, mi RNAs lsy-6, mir-273, and mir-71 function in asymmetric differentiation of two pairs of *C. elegans* sensory neurons, which will be discussed later. The hypothesis that differential expression of these genes between left and right sides of the cortex may be regulated by mi RNAs is plausible and worth further investigation. Although there are limited reports of the involvement of mi RNAs in the development of neuronal asymmetry in vertebrates, *C. elegans* has proved to be a powerful model organism to study lateralization of the nervous system due to its genetic amenability, as well as evidence of functional asymmetry having clear molecular correlates. The priming event allows boosting of lsy-6 mi RNA levels by the CHE-1 transcription factor in the ASER mother cell, leading to high levels of the mi RNA in the postmitotic ASER neuron. In the ASER neuron, lsy-6 directly represses an ASER promoting transcription factor COG-1, through physical binding of complementary bases in the cog-1 3'UTR. Thus downregulation of tir-1 expression by mir-71 is an efficient mechanism to inhibit calcium signaling in the cell becoming AWCON. mir-71 is regulated at both transcriptional and post-transcriptional levels in AWC.

Mentioned biomedical entities:

Genes: ASER, CHE-1 transcription factor, COG-1, DIE-1, LIN-14, LMO4, Myt1, Notch, TRIM32, die-1, lin-4, lsy-6, mir-273, mir-71, nsy-4, str-2, xrn-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ASER cell, ASER lineage, ASER cell, AWC cells, AWCOFF cell, AWCON cell, AWCON neuron, NIH, SVZ, TBX-37/38, TRIM32, Tanaka-Hino, bipolar cell, cell, cells, left, mitral cells, neural, neuron, neuronal, neurons, postmitotic ASER, postmitotic ASER neuron, progenitor, progenitor cell, retinal progenitor cells

Organisms: ASER, *C. elegans*, *C. elegans* nervous system, *Xenopus* retina, brain, children, humans, mice, olfactory, patients, paw, zebrafish, zebrafish brain, zebrafish olfactory bulb

Tissues: ASER lineage, ASER neuron, ASER precursor, ASER precursor cell, amphid wing C, lateral, nervous system, ventriculardorsal subventricular zone (

Pathways: N/A

Query Result 10/25**Cell-type specific sequencing of microRNAs from complex animal tissues**

Alberti, Chiara; Manzenreither, Raphael A.; Sowemimo, Ivica; Burkard, Thomas R.; Wang, Jingkui; Mahofsky, Katharina; Ameres, Stefan L.; Cochella, Luisa

Nature methods

Year: 2018

PMID: 29481550

doi: 10.1038/nmeth.4610

Citations: 0

Score: 6.28

Keywords hits: ASE: 57, che-1: 2, lsy-6: 5

Abstract:

MicroRNAs (miRNAs) play an essential role in the post-transcriptional regulation of animal development and physiology. However, in vivo studies aimed at linking miRNA-function to the biology of distinct cell types within complex tissues remain challenging, partly because in vivo miRNA-profiling methods lack cellular resolution. We report microRNome by methylation-dependent sequencing (mime-seq), an in vivo enzymatic small RNA-tagging approach that enables high-throughput sequencing of tissue- and cell-type-specific miRNAs in animals. The method combines cell-type-specific 3-terminal 2-O-methylation of animal miRNAs by a genetically encoded, plant-specific methyltransferase (HEN1), with chemoselective small RNA cloning and high-throughput sequencing. We show that mime-seq uncovers the miRNomes of specific cells within *C. elegans* and *Drosophila* at unprecedented specificity and sensitivity, enabling miRNA profiling with single-cell resolution in whole animals. Mime-seq overcomes current challenges in cell-type-specific small RNA profiling and provides novel entry points for understanding the function of miRNAs in spatially restricted physiological settings.

Summary:

1g) in the henn-1 background^{24,28}. This lower overall recovery was expected, as mi R strands are typically low in abundance because of their rapid decay upon loading of the partner mi R strand onto Argonaute^{3,31}. Our results suggest that directing Ath-HEN1 expression to specific cells should enable recovery of the most relevant mi RNAs expressed in a single cell out of whole *C. elegans* worms at a TPR of .88, and even in single neurons within fly brains at a TPR of .8, in both cases with very low FPRs. To test if Ath-HEN1-directed methylation followed by oxidation-resistant cloning preserves the relative quantity of mi RNAs, we compared the abundance of untreated *Drosophila* S2 mi RNAs to that of the *Drosophila* mi RNAs that are significantly enriched after oxidation, throughout the S2-Ath-HEN1/m ESC dilution series. In direct comparison with neuronal and intestinal Argonaute immunoprecipitations in *C. elegans*^{13,14}, mime-seq identified 3- and 2.6-fold more mi RNAs respectively, many of which we validated, including a mi RNA expressed in a single neuron out of the whole animal³⁴. Comparison of the mi RNomes obtained by mime-seq with transcriptional reporters for several *C. elegans* mi RNAs showed a high correspondence, both for enriched and depleted mi RNAs. Note, that one expressed mi RNA aligned to both the mouse and the *Drosophila* genome but was assigned to the *Drosophila* sample because mi R-184 is only spuriously expressed in m ESC. The Nextflow⁶⁶ pipeline developed for this analysis, as well as the R script used for further analysis can be found at: <https://gitlab.com/tburk/small-RNA-meth-Data-analysis> To test the enrichment of *Drosophila* mi RNAs in the titration experiment in. Wrote the paper and prepared the figures. The authors declare no competing financial interests.¹ Bartel DPMicro RNAs: target recognition and regulatory functions Cell²⁹136215233191673261.116/j. cell.29.1.2 PMC37948962 Aboobaker AATomancak PPatel NRubin GMLai ECDrosophila micro RNAs exhibit diverse spatial expression patterns during embryonic development Proc Natl Acad Sci USA²⁵121817182216337591.173/pnas.5882312 PMC1367963 Landgraf PA Mammalian micro RNA Expression Atlas Based on Small RNA Library Sequencing Cell²⁷129141141417647271.116/j. cell.27.4.4 PMC26812314 Martinez NJGenome-scale spatiotemporal analysis of Caenorhabditis elegans micro RNA promoter activity Genome Research²⁸1825215189812661.111/gr.8355.18 PMC25935835 Park CYA resource for the conditional ablation of micro RNAs in the mouse Cell Reports²¹213853911.116/j. celrep.212.2.8 PMC3345172257876 Wienholds EMicro RNA expression in zebrafish embryonic development Science²⁵3931311159199541.1126/science.11145197 Chen WQin CGGeneral hallmarks of micro RNAs in brain evolution and development RNA Biology²¹5127178267281.18/15476286.215.148954 PMC46158398 Abruzzi KChen XNagoshi EZadina ARosbash MRNA-seq profiling of small numbers of *Drosophila* neurons Meth Enzymol²¹5551369386256624651.116/bs. mie.214.1.259 Ristori ENicoli Smi RNAs expression profile in zebrafish developing vessels Methods Mol Biol²¹512141291525468611.17/978-1-4939-1462-371 Hackler LMasuda TOliver VFMerbs SLZack DJUse of laser capture microdissection for analysis of retinal m RNA/mi RNA expression and DNA methylation Methods Mol Biol²¹288428934226887151.17/978-1-61779-848-121 PMC415436111 Herai RRMicro RNA detection in long-term fixed tissue of cortical glutamatergic pyramidal neurons after targeted laser-capture neuroanatomical microdissection J Neurosci Methods²¹42357682249925731.116/j. jneumeth.214.6.28 PMC41583212 Juzenas SA comprehensive, cell specific micro RNA catalogue of human peripheral blood Nucleic Acids Research²¹74592993128934571.193/nar/gkx76 PMC576619213 Kudlow BAZhang LHan MSsystematic analysis of tissue-restricted mi RISCs reveals a broad role for micro RNAs in suppressing basal activity of the *C. elegans* pathogen response Molecular Cell²¹2465354122534241.116/j. molcel.212.3.11 PMC336553514 Than MTKudlow BAHan MFfunctional analysis of neuronal micro RNAs in Caenorhabditis elegans dauer formation by combinatorial genetics and Neuronal mi RISC immunoprecipitation PLo S Genet²¹39e13592238188741.1371/journal. pgen.13592 PMC36885215 He

MCell-type-based analysis of micro RNA profiles in the mouse brain Neuron212733548222437451.116/j. neuron.211.11.1 PMC32749416 Yu BMethylation as a crucial step in plant micro RNA biogenesis Science253793293515758541.1126/science.11713 PMC51373717 Ohara TThe 3 termini of mouse Piwi-interacting RNAs are 2-O-methylated Nat Struct Mol Biol271434935173846461.138/nsmb12218 Houwing SA role for Piwi and pi RNAs in germ cell maintenance and transposon silencing in Zebrafish Cell271296982174187871.116/j. cell.27.3.2619 Yang ZEBright YWYu BChen XHEN1 recognizes 21-24 nt small RNA duplexes and deposits a methyl group onto the 2' OH of the 3' terminal nucleotide Nucleic Acids Research263466767516449231.193/nar/gkj474 PMC13565332 Ghildiyal MEndogenous si RNAs derived from transposons and mRNAs in Drosophila somatic cells Science283217718118436771.1126/science.1157396 PMC295324121 Alefelder SPatel BKEckstein FIncorporation of terminal phosphorothioates into oligonucleotides Nucleic Acids Research1998264983498897767631.193/nar/26.21.4983 PMC14794522 Horwich MDThe Drosophila RNA methyltransferase, Dm Hen1, modifies germline pi RNAs and single-stranded si RNAs in RISC Curr Biol27171265127217646291.116/j. cub.27.6.323 Saito KPimet, the Drosophila homolog of HEN1, mediates 2-O-methylation of Piwi- interacting RNAs at their 3' ends Genes Development272116316817666381.111/gad.156367 PMC189946924 Billi ACThe Caenorhabditis elegans HEN1 ortholog, HENN-1, methylates and stabilizes select subclasses of germline small RNAs PLo S Genet2128e126172254811.1371/journal. pgen.12617 PMC3339525 Kamminga LMDifferential impact of the HEN1 homolog HENN-1 on 21 U and 26 G RNAs in the germline of Caenorhabditis elegans PLo S Genet2128e1272228297721.1371/journal. pgen.1272 PMC3457626 Montgomery TAPIWI associated si RNAs and pi RNAs specifically require the Caenorhabditis elegans HEN1 ortholog henn-1 PLo S Genet2128e12616225361581.1371/journal. pgen.12616 PMC333488127 Park WLi JSong RMessing JChen XCARPEL FACTORY, a Dicer Homolog, and HEN1, a Novel Protein, Act in micro RNA Metabolism in Arabidopsis thaliana Current Biology221214841495122256631.116/s96-9822117-5 PMC513737228 Frkjaer-Jensen CDavis MWailion MJorgensen EMImproved Mos1-mediated transgenesis in C. elegans Nat Meth21291171181.138/nmeth.1865 PMC3725292222918129 Love MIHuber WAnders SModerated estimation of fold change and dispersion for RNA-seq data with DESeq2 Genome Biol2141555255162811.1186/s1359-14-55-8 PMC432493 Leuschner PJFameres SLKuang SMartinez JCleavage of the si RNA passenger strand during RISC assembly in human cells EMBO Rep26731432164399951.138/sj. embor.74637 PMC145689231 Frstemann KHorwich MDWee LTomari YZamore PDDrosophila micro RNAs Are Sorted into Functionally Distinct Argonaute Complexes after Production by Dicer-1 Cell2713287297176629431.116/j. cell.27.5.56 PMC26861932 Stefanakis NCarrera IHobert ORegulatory Logic of Pan-Neuronal Gene Expression in C. elegans Neuron2158773375262911581.116/j. neuron.215.7.31 PMC454549833 Tursun BCochella LCarrera IHobert OA toolkit and robust pipeline for the generation of fosmid-based reporter genes in C. elegans PLo S ONE294e4625192592641.1371/journal. pone.4625 PMC26495534 Cochella LHobert OEmbryonic priming of a mi RNA locus predetermines postmitotic neuronal left/right asymmetry in C. elegans Cell2121511229124223211431.116/j. cell.212.1.49 PMC35291435 Drexel TMahofsky KLatham RZimmer MCochella LNeuron type-specific mi RNA represses two broadly expressed genes to modulate an avoidance behavior in C. elegans Genes Development21632422472768841.111/gad.28794.116 PMC56661136 Johnston RJJr Hobert OA micro RNA controlling left/right neuronal asymmetry in Caenorhabditis elegans Nature2342684584914685241.138/nature225537 Clark AMThe micro RNA mi R-124 controls gene expression in the sensory nervous system of Caenorhabditis elegans Nucleic Acids Research2138378379321765731.193/nar/gkq83 PMC288795638 Hsieh Y-WChang CChuang C-FThe micro RNA mir-71 inhibits calcium signaling by targeting the TIR-1/Sarm1 adaptor protein to control stochastic L/R neuronal asymmetry in C. elegans PLo S Genet2128e128642287621.1371/journal. pgen.12864 PMC34185739 Simon DJThe Micro RNA mi R-1 Regulates a MEF-2-Dependent Retrograde Signal at Neuromuscular Junctions Cell281339391518519331.116/j. cell.28.4.35 PMC25535664 Sokol NSAmbros VMesodermally expressed Drosophila micro RNA-1 is regulated by Twist and is required in muscles during larval growth Genes Development251923432354161663731.111/gad.135615 PMC1244341 Cerro-Herreros EFernandez-Costa JMSabater-Arcis MLlamusi BARtero RDerepressing muscleblind expression by mi RNA sponges ameliorates myotonic dystrophy-like phenotypes in Drosophila Sci Rep2161131.138/srep3623278516 PMC5924642 Viswanathan SRDaley GQGregory RISElective blockade of micro RNA processing by Lin28 Science283297118292371.1126/science.11544 PMC336849943 Jannot GBoisvert M-ELBanville IHSimard MJTwo molecular features contribute to the Argonaute specificity for the micro RNA and RNAi pathways in C. elegans RNA2814829835183677181.1261/rna.9198 PMC232735744 Steiner FAStructural features of small RNA precursors determine Argonaute loading in Caenorhabditis elegans Nat Struct Mol Biol2714927933178911481.138/nsmb13845 Tomari YDu TZamore PDSorting of Drosophila small silencing RNAs Cell271329938176629441.116/j. cell.27.5.57 PMC28415546 Okamura KLi NLai ECDistinct mechanisms for micro RNA strand selection by Drosophila Argonautes Molecular Cell2936431444199172511.116/j. molcel.29.9.27 PMC27857947 Czech BHierarchical rules for Argonaute loading in Drosophila Molecular Cell2936445456199172521.116/j. molcel.29.9.28 PMC279532548 Ghildiyal MXu JSeitz HWeng ZZamore PDSorting of Drosophila small silencing RNAs partitions micro RNA strands into the RNA interference pathway RNA21164356199176351.1261/rna.197291 PMC2823649 Corra RLSteiner FABerezikov EKetting RFMicro RNADirected si RNA Biogenesis in Caenorhabditis elegans PLo S Genet216e19323867451.1371/journal. pgen.193 PMC28515715 Alberti CPlaceholder for protocol exchange reference51 Brenner SThe genetics of Caenorhabditis elegans Genetics197477719443664761.193/genetics/77.1.71 PMC12131252 Sulston JEHodgkin JThe Nematode Caenorhabditis elegans Cold Spring Harbor Laboratory Press19885876653 Redemann SCodon adaptation-based control of protein expression in C. elegans Nat Meth2118252521.138/nmeth.15652127874354 Etchberger JFThe molecular signature and cis-regulatory architecture of a C. elegans gustatory

neuron Genes Development27211653167417666431.111/gad.15617 PMC189947455 Reimo-Pinto MMUridylation of RNA Hairpins by Tailor Confines the Emergence of Micro RNAs in Drosophila Molecular Cell2155923216261451761.116/j. molcel.215.5.33 PMC45183956 Ameres SLTarget RNA-directed trimming and tailing of small silencing RNAs Science213281534153925587121.1126/science.118758 PMC29298557 Gokcezade JSiensi GDuchek PEfficient CRISPR/Cas9 plasmids for rapid and versatile genome editing in Drosophila G3: GenesGenomesGenetics214422792282252367341.1534/g3.114.14126 PMC423255358 Markstein MPitsouli CVillalta CCelniker SEPerrimon NExploiting position effects and the gypsy retrovirus insulator to engineer precisely expressed transgenes Nature Genetics284476483183111411.138/ng.11 PMC23326159 Bryantsev ALBaker PWLovato TLJaramillo MSCripps RMDifferential requirements for Myocyte Enhancer Factor-2 during adult myogenesis in Drosophila Developmental Biology212361191272287921.116/j. ydbio.211.9.31 PMC32465366 Klein PRet rescues mitochondrial morphology and muscle degeneration of Drosophila Pink1 mutants The EMBO Journal21433341355244731491.12/embj.2128429 PMC39836861 Bassett ARTibbit CPonting CPLiu J-LMutagenesis and homologous recombination in Drosophila cell lines using CRISPR/Cas9 Biol Open21434249243261861.1242/bio.213712 PMC389215962 Schindelin JFiji: an open-source platform for biological-image analysis Nat Meth21296766821.138/nmeth.219 PMC38558442274377263 Han BWHung J-HWeng ZZamore PDAmers SLThe 3-to-5 Exoribonuclease Nibbler Shapes the 3 Ends of Micro RNAs Bound to Drosophila Argonaute1 Current Biology211211878188722552931.116/j. cub.211.9.34 PMC323649964 Martin MCutadapt removes adapter sequences from high-throughput sequencing reads EMBnetjournal21117165 Anders SPyl PTHuber WHTSeqa Python framework to work with high-throughput sequencing data Bioinformatics21531166169252671.193/bioinformatics/btu638 PMC42879566 Di Tommaso PNextflow enables reproducible computational workflows Nat Biotechnol21735316319283983111.138/nbt.38267 Micallef LRodgers Peuler APE: Drawing Area-Proportional 3-Venn Diagrams Using Ellipses PLo S ONE2149e1171725328251.1371/journal. pone.11717 PMC412485 Editorial summary Mime-seq achieves cell-type specific, methylation-based, micro RNA tagging and sequencing, to uncover cell-specific micro RNomes in C. elegans and Drosophila.

Mentioned biomedical entities:

Genes: Actin, Actin5C, Argonaute, Cas9, Dicer, Gal4, HEN1, HEN1 homolog, HENN-1, Hen1, Pink1, Piwi, RDE-1, Tubulin, Twist, Unc, attB, lsy-6, miR-252, miR-277, rab-3, rgef-1, unc-31

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: 2/558 cells, ASE, Ath-HEN1-CM-fwd-3, C, Cell, Cells, Clone, Drosophila S2 cells, Drosophila cell lines, Drosophila cells, E., E. coli BL21, S2 cell, S2 cells, S2-Ath-HEN1/mESC, Schneider, Sephadex G-25 spin column, Sorted, Splenic B-cells, Suppl, TPR, X63- Ag8.653 mouse myeloma cells, b., cell, cell-culture, cell-type, cells, cellular, clonal, clone M2, clones, cortical glutamatergic, fluorescence-activated cell, fly, germ cell, <https://gitlab.com/tburk/smallRNA-meth>C. elegans strains, laser-capture, laser-capture neuroanatomical microdissectionJ Neurosci Methods2142357682249925731.116/j. jneumeth.214.6.28PMC41583212JuzenasSA, lines, mESC, neuron, neuronal, neurons, ng/l elt-2::DsRed, non-Unc, non-neuronal cells, periodate-induced, postmitotic neuronal, round-shaped clones, secretory cells, single-cell clones, wells, wild-type, 1/3 cells, 2/3

Organisms: C., C. elegans, C. elegans genome, C. elegans lines, C. elegans nervous system, Mus musculus, brain, human peripheral, intestine, mouse, mouse embryonic stem cells, muscles, nervous, organs, pharynx, zebrafish

Tissues: F1, body-wall muscle, body-wall muscles, cuticle, layer, muscle, muscles, pads, seed, tissue, tissues, tube

Pathways: N/A

Query Result 11/25**MicroRNAs: Not Fine-Tuners but Key Regulators of Neuronal Development and Function**

Davis, Gregory M.; Haas, Matilda A.; Pocock, Roger

Frontiers in Neurology

Year: 2015

PMID: 26635721

doi: 10.3389/fneur.2015.00245

Citations: 0

Score: 5.57

Keywords hits: ASE: 63, che-1: 1, lsy-6: 7

Abstract:

MicroRNAs (miRNAs) are a class of short non-coding RNAs that operate as prominent post-transcriptional regulators of eukaryotic gene expression. miRNAs are abundantly expressed in the brain of most animals and exert diverse roles. The anatomical and functional complexity of the brain requires the precise coordination of multilayered gene regulatory networks. The flexibility, speed, and reversibility of miRNA function provide precise temporal and spatial gene regulatory capabilities that are crucial for the correct functioning of the brain. Studies have shown that the underlying molecular mechanisms controlled by miRNAs in the nervous systems of invertebrate and vertebrate models are remarkably conserved in humans. We endeavor to provide insight into the roles of miRNAs in the nervous systems of these model organisms and discuss how such information may be used to inform regarding diseases of the human brain.

Summary:

Prior to the discovery of the mi RNA pathway, the lin-14 gene in *Caenorhabditis elegans* was shown to be regulated by a 22-nucleotide partially complementary strand of RNA called lin-4. Mir-1 also targets synaptic proteins neuroligin and neurexin, which in humans are two synaptic proteins that have been linked with defects in synaptic function associated with autism spectrum disorders. For example, control of neuronal progenitor proliferation is fine-tuned by the highly conserved mi RNA, mi R-124, which has been shown in various organisms to regulate neuronal stem cells. Deletion of Dicer leads to neuron migration defects and impaired cellular differentiation, as well as cell fate changes and cortical lamination defects. Conditional deletion of Dicer from the embryonic day 8 telencephalon causes a loss of radial glial progenitor markers, including nestin, Sox9, and Erb B2, which then results in an increase in basal progenitors and postmitotic neurons. Defects associated not only with the expression of key mi RNAs, but also at a genetic level have been implicated in schizophrenia.

Mentioned biomedical entities:

Genes: AGO, ASCL1, ASER, Akt, Argonaute, CHE-1, COG-1, CamkII, DAF-16, Dgcr8, Dicer, Drosha, EGFR, Emx1, ErbB2, FMR1, FOXO, Fgf, Fgf8, Fmr1, Foxg1, Foxp2, Gsh2, Hes1, LIN-14, LIN-41, Let-7, Lmx1b, MAP2, MECP2, MEF-2, MLN51, MYT1L, MeCP2, NEUROD2, NRXN1, Notch, Nrg, Pasha, Phf6, RNA, RhoG, SHANK3, SOX-2, SQV-5, Sox9, Syt1, TLX, TRIM, TRIM32, UNC-4, Wnt, XPO5, ana, chinmo, dicer, dme, fgfr1, her5, let-7, lin-14, lin-28, lin-4, lin-41, lsy-6, miR-124, miR-125, miR-125b, miR-137, miR-228, miR-289, miR-7, miR-9, miR-92b, miR-9a, mir-71, mir-79, pitx3, zic5

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: AB lineage, ASEL neuron, Cell, Forssman, MHB, NMJ, Nervous SystemThe human brain, Neuron, Nsf, SOP cells, SQV-5, axon, basal progenitors, cell, cell lines, cell-specific, cells, cellular, embryos, ensheathing, epidermal cells, epithelial cells, foreskin fibroblasts, glial cell, glial cells, glial progenitor, mammalian cells, neural cells, neural crest cells, neural progenitor, neural progenitor cell, neural progenitor cells, neural progenitors, neural stem cell, neural stem cells, neuroblast, neuroepithelial, neuron, neuronal, neuronal cells, neuronal lineage, neuronal progenitor, neuronal stem cells, neurons, non-neuronal, non-neuronal cells, nsy-4/claudin, presynaptic cell, progenitor, progenitor cells, retinal progenitor cells, stem cell

Organisms: *C. elegans*, *Homo sapiens*, *Mus musculus*, body, brain, children, eye, hindbrain, human, human brain, human embryonic stem cells, human nervous system, human pluripotent stem cells, human stem cells, human tissues, humans, intestine, mice, midbrain, midbrain-hindbrain, mouse, mouse brain, mouse brains, mouse cortex, mushroom body, nervous, organ, organs, patients, synaptic, zebrafish, zebrafish embryos, zebrafish hindbrain

Tissues: epidermis, epithelial, muscle, nervous, nervous system, nervous system to, neural crest, neural tube, neuromuscular junction, olfactory, optic lobe, neurepithelium, organ precursor, placodes, tissue

Pathways: N/A

Query Result 12/25**MicroRNA dysregulation in psychiatric disease**

Miller, Brooke H.; Wahlestedt, Claes

Brain research

Year: 2010

PMID: 20303342

doi: 10.1016/j.brainres.2010.03.035

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

MicroRNAs (miRNAs) are small regulatory RNAs that individually regulate up to several hundred genes, and collectively may regulate as much as two-thirds of the transcriptome. Recent evidence supports a role for miRNA dysregulation in psychiatric and neurological disorders, including schizophrenia, bipolar disorder, and autism. Small changes in miRNA expression can fine-tune the expression of multiple genes within a biological network, suggesting that miRNA dysregulation may underlie many of the molecular changes observed in psychiatric disease, and that therapeutic regulation of miRNA levels may represent a novel treatment option.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 13/25Notch-dependent induction of left/right asymmetry in *C. elegans* interneurons and motoneurons

Bertrand, Vincent; Bisso, Paul; Poole, Richard J.; Hobert, Oliver

Current biology : CB

Year: 2011

PMID: 21737278

doi: 10.1016/j.cub.2011.06.016

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Although invertebrate and vertebrate nervous systems are largely bilateral symmetric on a structural level, they display striking degrees of functional left/right (L/R) asymmetry. In *Caenorhabditis elegans* two structurally symmetric pairs of sensory neurons, ASE and AWC display distinct types of functional L/R asymmetry (stereotyped vs. stochastic asymmetry), which are controlled by distinct molecular mechanisms 13. Beyond these two cases, the extent of neuronal asymmetry in the *C. elegans* nervous system and the types and diversities of mechanisms that control other potential asymmetries remain to be determined. Here, we report that the Beta3/Olig-type bHLH transcription factor *hlh-16* is L/R asymmetrically expressed in several distinct, otherwise bilaterally symmetric interneuron and motoneuron pairs. These neuron pairs are part of a known circuit controlling *C. elegans* navigation. We find that *hlh-16* asymmetry is generated during gastrulation by an asymmetric LAG-2/Delta signal originating from the mesoderm. This signal promotes *hlh-16* expression in neurons on the left side through direct binding of the Notch effector LAG-1/Su(H)/CBF to a cis-regulatory element in the *hlh-16* locus. Through the removal of *hlh-16* we reveal an unanticipated asymmetry in the ability of the axons of the AIY interneurons to extend into the nerve ring, with the left AIY axon requiring elevated *hlh-16* expression for correct extension. Taken together, our study suggests that the extent of molecular L/R asymmetry in the *C. elegans* nervous system is broader than previously anticipated, it establishes a novel signaling mechanism that crosses germ layers to diversify bilaterally symmetric neuronal lineages, and reveals L/R asymmetric control of axonal outgrowth of bilaterally symmetric neurons.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 14/25

Maintenance of neuronal identity in *C. elegans* and beyond: lessons from transcription and chromatin factors

Destain, Honorine; Prahlad, Manasa; Kratsios, Paschalis

Seminars in cell developmental biology

Year: 2024

PMID: 37438210

doi: 10.1016/j.semcd.2023.07.001

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Neurons are remarkably long-lived, non-dividing cells that must maintain their functional features (e.g., electrical properties, chemical signaling) for extended periods of time decades in humans. How neurons accomplish this incredible feat is poorly understood. Here, we review recent advances, primarily in the nematode *C. elegans*, that have enhanced our understanding of the molecular mechanisms that enable post-mitotic neurons to maintain their functionality across different life stages. We begin with terminal selectors - transcription factors necessary for the establishment and maintenance of neuronal identity. We highlight new findings on five terminal selectors (CHE-1 Glass, UNC-3 Collier/Ebf1-4, LIN-39 Scr/Dfd/Hox4-5, UNC-86 Acj6/Brn3ac, AST-1 Etv1/ER81) from different transcription factor families (ZNF, COE, HOX, POU, ETS). We compare the functions of these factors in specific neuron types of *C. elegans* with the actions of their orthologs in other invertebrate (*D. melanogaster*) and vertebrate (*M. musculus*) systems, highlighting remarkable functional conservation. Finally, we reflect on recent findings implicating chromatin-modifying proteins, such as histone methyltransferases and Polycomb proteins, in the control of neuronal terminal identity. Altogether, these new studies on transcription factors and chromatin modifiers not only shed light on the fundamental problem of neuronal identity maintenance, but also outline mechanistic principles of gene regulation that may operate in other long-lived, post-mitotic cell types.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 15/25**Recent Molecular Genetic Explorations of *Caenorhabditis elegans* MicroRNAs**

Ambros, Victor; Ruvkun, Gary

Genetics

Year: 2018

PMID: 29967059

doi: 10.1534/genetics.118.300291

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

MicroRNAs are small, noncoding RNAs that regulate gene expression at the post-transcriptional level in essentially all aspects of *Caenorhabditis elegans* biology. More than 14 genes that encode microRNAs in *C. elegans* regulate development, behavior, metabolism, and responses to physiological and environmental changes. Genetic analysis of *C. elegans* microRNA genes continues to enhance our fundamental understanding of how microRNAs are integrated into broader gene regulatory networks to control diverse biological processes, including growth, cell division, cell fate determination, behavior, longevity, and stress responses. As many of these microRNA sequences and the related processing machinery are conserved over nearly a billion years of animal phylogeny, the assignment of their functions via worm genetics may inform the functions of their orthologs in other animals, including humans. In vivo investigations are especially important for microRNAs because in silico extrapolation of their functions using mRNA target prediction programs can easily assign microRNAs to incorrect genetic pathways. At this mezzanine level of microRNA bioinformatic sophistication, genetic analysis continues to be the gold standard for pathway assignments.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 16/25**Autophagy in *C. elegans* development**

Palmisano, Nicholas J.; Melndez, Alicia

Developmental biology

Year: 2018

PMID: 29709599

doi: 10.1016/j.ydbio.2018.04.009

Citations: 24

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Autophagy involves the sequestration of cytoplasmic contents in a double-membrane structure referred to as the autophagosome and the degradation of its contents upon delivery to lysosomes. Autophagy activity has a role in multiple biological processes during the development of the nematode *Caenorhabditis elegans*. Basal levels of autophagy are required to remove aggregate prone proteins, paternal mitochondria, and spermatid-specific membranous organelles. During larval development, autophagy is required for the remodeling that occurs during dauer development, and autophagy can selectively degrade components of the miRNA-induced silencing complex, and modulate miRNA-mediated silencing. Basal levels of autophagy are important in synapse formation and in the germ line, to promote the proliferation of proliferating stem cells. Autophagy activity is also required for the efficient removal of apoptotic cell corpses by promoting phagosome maturation. Finally, autophagy is also involved in lipid homeostasis and in the aging process. In this review, we first describe the molecular complexes involved in the process of autophagy, its regulation, and mechanisms for cargo recognition. In the second section, we discuss the developmental contexts where autophagy has been shown to be important. Studies in *C. elegans* provide valuable insights into the physiological relevance of this process during metazoan development.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 17/25**Integration of Spatial and Temporal Patterning in the Invertebrate and Vertebrate Nervous System**

Chen, Yen-Chung; Konstantinides, Nikolaos

Frontiers in Neuroscience

Year: 2022

PMID: 35392413

doi: 10.3389/fnins.2022.854422

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

The nervous system is one of the most sophisticated animal tissues, consisting of thousands of interconnected cell types. How the nervous system develops its diversity from a few neural stem cells remains a challenging question. Spatial and temporal patterning mechanisms provide an efficient model through which diversity can be generated. The molecular mechanism of spatiotemporal patterning has been studied extensively in *Drosophila melanogaster*, where distinct sets of transcription factors define the spatial domains and temporal windows that give rise to different cell types. Similarly, in vertebrates, spatial domains defined by transcription factors produce different types of neurons in the brain and neural tube. At the same time, different cortical neuronal types are generated within the same cell lineage with a specific birth order. However, we still do not understand how the orthogonal information of spatial and temporal patterning is integrated into the progenitor and post-mitotic cells to combinatorially give rise to different neurons. In this review, after introducing spatial and temporal patterning in *Drosophila* and mice, we discuss possible mechanisms that neural progenitors may use to integrate spatial and temporal information. We finally review the functional implications of spatial and temporal patterning and conclude envisaging how small alterations of these mechanisms can lead to the evolution of new neuronal cell types.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 18/25**Mir-17-3p Controls Spinal Neural Progenitor Patterning by Regulating Olig2/Irx3 Cross-repressive Loop**

Chen, Jun-An; Huang, Yuan-Ping; Mazzoni, Esteban O.; Tan, G. Christopher; Zavadil, Jiri; Wichterle, Hynek

Neuron

Year: 2011

PMID: 21338882

doi: 10.1016/j.neuron.2011.01.014

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Neural patterning relies on transcriptional cross-repressive interactions that ensure unequivocal assignment of neural progenitor identity to proliferating cells. Progenitors of spinal motor neurons (pMN) and V2 interneurons (p2) are specified by a pair of cross-repressive transcription factors Olig2 and Irx3. Lineage tracing revealed that many p2 progenitors transiently express the pMN marker Olig2 during spinal cord development. Here we demonstrate that the repression of Olig2 in p2 domain is controlled by mir-17-3p microRNA-mediated silencing of Olig2 mRNA. Mice lacking all microRNAs or just the mir-1792 cluster manifest a dorsal shift in pMN/p2 boundary and impairment in the production of V2 interneurons. Our findings suggest that microRNA-mediated repression of Olig2 mRNA plays a critical role during the patterning of ventral spinal progenitor domains by shifting the balance of cross-repressive interactions between Olig2 and Irx3 transcription factors.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 19/25

A growing molecular toolbox for the functional analysis of microRNAs in *Caenorhabditis elegans*

Jo, Jeanyoung; Esquela-Kerscher, Aurora

Briefings in Functional Genomics

Year: 2011

PMID: 21624898

doi: 10.1093/bfpg/elr012

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

With the growing number of microRNAs (miRNAs) being identified each year, more innovative molecular tools are required to efficiently characterize these small RNAs in living animal systems. *Caenorhabditis elegans* is a powerful model to study how miRNAs regulate gene expression and control diverse biological processes during development and in the adult. Genetic strategies such as large-scale miRNA deletion studies in nematodes have been used with limited success since the majority of miRNA genes do not exhibit phenotypes when individually mutated. Recent work has indicated that miRNAs function in complex regulatory networks with other small RNAs and protein-coding genes, and therefore the challenge will be to uncover these functional redundancies. The use of miRNA inhibitors such as synthetic antisense 2'-O-methyl oligoribonucleotides is emerging as a promising in vivo approach to dissect out the intricacies of miRNA regulation.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 20/25**Stochasticity and Cell Fate**

Losick, Richard; Desplan, Claude

Science (New York, N.Y.)

Year: 2008

PMID: 18388284

doi: 10.1126/science.1147888

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Fundamental to living cells is the capacity to differentiate into subtypes with specialized attributes. Understanding the way cells acquire their fates is a major challenge in developmental biology. How cells adopt a particular fate is usually thought of as being deterministic, and in the large majority of cases it is. That is, cells acquire their fate by virtue of their lineage or their proximity to an inductive signal from another cell. In some cases, however, and in organisms ranging from bacteria to humans, cells choose one or another pathway of differentiation stochastically without apparent regard to environment or history. Stochasticity has important mechanistic requirements as we discuss. We will also speculate on why stochasticity is advantageous, and even critical in some circumstances, to the individual, the colony, or the species.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 21/25

LSY-2 is essential for maintaining the germ-soma distinction in *C. elegans*

Lin, Long; Li, Yuping; Yan, Libo; Zhang, Gangming; Zhao, Yu; Zhang, Hong

Protein Cell

Year: 2015

PMID: 26050091

doi: 10.1007/s13238-015-0173-1

Citations: 0

Score: 0.00

Keywords hits: ASE: 38, che-1: , lsy-6: 2

Abstract:

The mechanisms that specify and maintain the characteristics of germ cells during animal development are poorly understood. In this study, we demonstrated that loss of function of the zinc-finger gene *lsy-2* results in various somatic cells adopting germ cells characteristics, including expression of germline-specific P granules, enhanced RNAi activity and transgene silencing. The soma to germ transformation in *lsy-2* mutants requires the activities of multiple chromatin remodeling complexes, including the MES-4 complex and the ISW-1 complex. The distinct germline-specific features in somatic cells and the gene expression profile indicate that LSY-2 acts in the Mec complex in this process. Our study demonstrated that *lsy-2* functions in the maintenance of the soma-germ distinction.

Summary:

In autophagy mutants, PGL-1 and PGL-3 colocalize and accumulate into aggregates in somatic cells. Establishment of somatic and germ precursor cells during early embryogenesis in *C. elegans* requires maternally distributed proteins, including PIE-1, MEX-1, MEX-3, and POS-1, which are also associated with P granules during the first several cell divisions and disappear in the daughters of the germ precursor P4 cells. Percentage of wild-type and *lsy-2* mutant animals that were arrested at the L3 stage after feeding with RNAi bacterial clones of *his-44* and *cel-1*. Therefore, somatic cells in *lsy-2* mutants display characteristics of germ cells. *lsy-2* functions similar to Mec class genes in repressing expression of germline genes in somatic cells. Based on phenotypic differences and the different profile of misexpressed RNAi factors and P-granule components in somatic cells, Syn Muv B genes can be divided into three distinct classes: a DRM core complex, a SUMO-recruited Mec complex, and a Syn Muv B heterochromatin complex. In addition to components of the *lin-35* Rb pathway, including *lin-35*, *lin-53*, *hpl-2*, *lin-9*, *dpl-1* and *lin-52*, and the Mec complex component *mep-1*, we identified that inactivation of the zinc-finger transcription factor *lsy-2* and components of the sumoylation pathway, including *smo-1*, *ubc-9*, and *uba-2*, caused ectopic accumulation of GFP::PGL-1 granules in somatic cells. Loss of function of *lsy-2* caused ectopic expression of GFP::PGL-1 in various somatic cells, including hypodermal cells and intestinal cells. To determine whether endogenous P-granule components were also ectopically expressed, we stained *lsy-2* mutant animals with anti-PGL-1 and anti-GLH-1 antibodies and found that endogenous PGL-1 and GLH-1 were also ectopically expressed and formed perinuclear granules in somatic cells (Fig).

Mentioned biomedical entities:

Genes: ASER, B, GLH-1, ISW1, ISWI, L1, LET-418, LIN-15B, MES-4, MEX-1, P4, PGL-1, PGL-3, PIE-1, Rb, SWR1, ZK1127.3, *cel-1*, *efl-1*, *gfl-1*, *glh-1*, *gon-14*, *his-44*, histone, *lim-6*, *lin-15A*, *lin-15B*, *lin-35*, *lin-49*, *lin-53*, *lin-59*, *lin-65*, *lin-9*, *lsy-6*, *nurf-1*, *sago-2*, *ubc-9*

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ASE cell, ASER cell, AD, BD, CE, *E. coli* BL21, F and G, GFP::PGL-1, Germ, Germ cells, INTRODUCTIONGerm cells, LAG-2::GFP (I)Germ cells, NIH, POS-1, SCM::GFP, bsomatic cells, bacterial clones, blastomere, blastomeres, cell, cell-type, cells, clone, clones, embryos, germ, germ cell, germ cell-specific, germ cells, germ precursor P4 cells, germ precursor cells, germ precursor cells Z2, germline cell, germline cells, hypodermal cells, intestinal cells, intestine, *let-418*, *lin-61*, lines, *lsy-2*, neurons, primordial germ cell P4, seam cell, seam cells, soma-germ, somatic, somatic cell, somatic cells, somatic founder cell, sperm, vulva cell, wild-type

Organisms: *C.*, *C. elegans*, H and I, L1 larvae, human, intestine, *lsy-2* somatic cells, rabbits, rat

Tissues: ASER neurons, F)

Pathways: N/A

Query Result 22/25**A Toolkit and Robust Pipeline for the Generation of Fosmid-Based Reporter Genes in *C. elegans*, bFosmid Recombineering**

Tursun, Baris; Cochella, Luisa; Carrera, Ins; Hobert, Oliver; Hart, Anne C.

PLoS ONE

Year: 2009 PMID: 19259264 doi: 10.1371/journal.pone.0004625 Citations: 0 Score: 0.00

Keywords hits: ASE: 26, che-1: 4, lsy-6:

Abstract:

Engineering fluorescent proteins into large genomic clones, contained within BACs or fosmid vectors, is a tool to visualize and study spatiotemporal gene expression patterns in transgenic animals. Because these reporters cover large genomic regions, they most likely capture all cis-regulatory information and can therefore be expected to recapitulate all aspects of endogenous gene expression. Inserting tags at the target gene locus contained within genomic clones by homologous recombination (recombineering) represents the most straightforward method to generate these reporters. In this methodology paper, we describe a simple and robust pipeline for recombineering of fosmids, which we apply to generate reporter constructs in the nematode *C. elegans*, whose genome is almost entirely covered in an available fosmid library. We have generated a toolkit that allows for insertion of fluorescent proteins (GFP, YFP, CFP, VENUS, mCherry) and affinity tags at specific target sites within fosmid clones in a virtually seamless manner. Our new pipeline is less complex and, in our hands, works more robustly than previously described recombineering strategies to generate reporter fusions for *C. elegans* expression studies. Furthermore, our toolkit provides a novel recombineering cassette which inserts a SL2-spliced intercistronic region between the gene of interest and the fluorescent protein, thus creating a reporter controlled by all 5 and 3 cis-acting regulatory elements of the examined gene without the direct translational fusion between the two. With this configuration, the onset of expression and tissue specificity of secreted, sub-cellular compartmentalized or short-lived gene products can be easily detected. We describe other applications of fosmid recombineering as well. The simplicity, speed and robustness of the recombineering pipeline described here should prompt the routine use of this strategy for expression studies in *C. elegans*.

Summary:

First, using Red recombinase, a cassette containing both the tag of interest and a selectable marker flanked by FRT/Flp sites is inserted by homologous recombination at the position of interest. 2). The reporter tag of choice is introduced in the form of a cassette also containing the selectable marker gal K flanked by FRT sites, which will later be targeted by the Flp recombinase (Fig. Previous reporter gene analysis with a 1 Kbp reporter construct indicated expression of the gene in ASE, as expected, but also in several other neuron types 22. The genomic fragment is fused by PCR to this cassette generating a Kan cassette flanked by the homology to one end of p CC1 FOS and the genomic extension fragment. P BALU-ext was used to amplify the kanamycin resistance gene flanked by sequences homologous to the fosmid backbone p CC1 FOS and to the genomic fragment of the gcy-22 gene that is missing from the fosmid.

Mentioned biomedical entities:

Genes: B, FgF, Kan, cfp, cog-1, galK, gcy-22, unc-54

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: BACs, FRT, SW15 E. coli, backbone.(.64 MB TIF)Click, cells, cellular, clone, clones, colonies, electrocompetent cells, gcy-22, neuron, non-selectable, sequencing.(.55 MB TIF)Click, unc-119

Organisms: *C. briggsae* unc-119, *C. elegans*, *C. elegans* unc-54 3 UTR, *E. coli*, Hobert lab, animals

Tissues: FRT-galK module, tissue

Pathways: N/A

Query Result 23/25Dissection of Genetic Pathways in *C. elegans*

Wang, Zheng; Sherwood, David R.

Methods in cell biology

Year: 2011 PMID: 22118276 doi: 10.1016/B978-0-12-544172-8.000050 Citations: 0 Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

With unique genetic and cell biological strengths, *C. elegans* has emerged as a powerful model system for studying many biological processes. These processes are regulated by complex genetic networks consisting of arrays of genes. Identifying those genes and organizing them into genetic pathways are two major steps towards understanding the mechanisms that regulate biological events. Forward genetic screens with various designs are a traditional approach for identifying candidate genes. The completion of the genome sequencing in *C. elegans* and the advent of high-throughput experimental techniques have led to the development of two additional powerful approaches: functional genomics and systems biology. Genes that are identified by all these approaches can be ordered into interacting pathways through a variety of strategies, involving genetics, cell biology, biochemistry and functional genomics, to gain a complete understanding of how gene regulatory networks control a particular biological process. The aim of this review is to provide an overview of the approaches available to identify and construct the genetic pathways using *C. elegans*.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 24/25

Cis-regulatory mechanisms of left/right asymmetric neuron-subtype specification in *C. elegans*

Etchberger, John F.; Flowers, Eileen B.; Poole, Richard J.; Bashllari, Enkelejda; Hobert, Oliver

Development (Cambridge, England)

Year: 2009

PMID: 19060335

doi: 10.1242/dev.030064

Citations: 0

Score: 0.00

Keywords hits: ASE: , che-1: , lsy-6:

Abstract:

Anatomically and functionally defined neuron types are sometimes further classified into individual subtypes based on unique functional or molecular properties. To better understand how developmental programs controlling neuron type specification are mechanistically linked to programs controlling neuronal subtype specification, we have analyzed a neuronal subtype specification program that occurs across the left/right axis in the nervous system of the nematode *C. elegans*. A terminal selector transcription factor, CHE-1, is required for the specification of the ASE neuron class, and a gene regulatory feedback loop of transcription factors and miRNAs is required to diversify the two ASE neurons into an asymmetric left and right subtype (ASEL and ASER). However, the link between the CHE-1-dependent ASE neuron class specification and the ensuing left-right subtype specification program is poorly understood. We show here that CHE-1 has genetically separable functions in controlling bilaterally symmetric ASE neuron class specification and the ensuing left-right subtype specification program. Both neuron class specification and asymmetric subclass specification depend on CHE-1-binding sites (ASE motifs) in symmetrically and asymmetrically expressed target genes, but in the case of asymmetrically expressed target genes, the activity of the ASE motif is modulated through a diverse set of additional cis-regulatory elements. Depending on the target gene, these cis-regulatory elements either promote or inhibit the activity of CHE-1. The activity of these L/R asymmetric cis-regulatory elements is indirectly controlled by che-1 itself, revealing a feed-forward loop configuration in which che-1 restricts its own activity. Relative binding affinity of CHE-1 to ASE motifs also depends on whether a gene is expressed bilaterally or in a left/right asymmetric manner. Our analysis provides insights into the molecular mechanisms of neuronal subtype specification, demonstrating that the activity of a neuron type-specific selector gene is modulated by a variety of distinct means to diversify individual neuron classes into specific subclasses. It also suggests that feed-forward loop motifs may be a prominent feature of neuronal diversification events.

Summary:

N/A

Mentioned biomedical entities:

Genes: N/A

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: N/A

Organisms: N/A

Tissues: N/A

Pathways: N/A

Query Result 25/25

Poly(A)-binding proteins are required for microRNA-mediated silencing and to promote target deadenylation in *C. elegans*

Flamand, Mathieu N.; Wu, Edlyn; Vashisht, Ajay; Jannot, Guillaume; Keiper, Brett D.; Simard, Martin J.; Wohlschlegel, James; Duchaine, Thomas F.

Nucleic Acids Research

Year: 2016

PMID: 27095199

doi: 10.1093/nar/gkw276

Citations: 0

Score: 0.00

Keywords hits: ASE: 45, che-1: , lsy-6: 15

Abstract:

Cytoplasmic poly(A)-binding proteins (PABPs) link mRNA 3 termini to translation initiation factors, but they also play key roles in mRNA regulation and decay. Reports from mice, zebrafish and *Drosophila* further involved PABPs in microRNA (miRNA)-mediated silencing, but through seemingly distinct mechanisms. Here, we implicate the two *Caenorhabditis elegans* PABPs (PAB-1 and PAB-2) in miRNA-mediated silencing, and elucidate their mechanisms of action using concerted genetics, protein interaction analyses, and cell-free assays. We find that *C. elegans* PABPs are required for miRNA-mediated silencing in embryonic and larval developmental stages, where they act through a multi-faceted mechanism. Depletion of PAB-1 and PAB-2 results in loss of both poly(A)-dependent and -independent translational silencing. PABPs accelerate miRNA-mediated deadenylation, but this contribution can be modulated by 3UTR sequences. While greater distances with the poly(A) tail exacerbate dependency on PABP for deadenylation, more potent miRNA-binding sites partially suppress this effect. Our results refine the roles of PABPs in miRNA-mediated silencing and support a model wherein they enable miRNA-binding sites by looping the 3UTR poly(A) tail to the bound miRISC and deadenylase.

Summary:

The RL-6-mi R-52-p86 reporter was subjected to a translational repression assay in GST- or GST-PAIP2-treated extracts or in presence of 5 M soluble GST or GST-PAIP2. To further probe the mechanistic role of PAB-1 and PAB-2, we investigated whether the presence of a poly tail is required for miRNA-mediated silencing and if PABPs require a poly tail for their role in miRNA-mediated silencing. This result indicates that at least part of miRNA-mediated silencing in *C. elegans* embryo is independent of deadenylation or the presence of a poly tail. PAB-1 and PAB-2 are required for miRNA-mediated silencing independently of the poly tail. Together, these results show that the poly tail is not required for all miRNA-mediated silencing in *C. elegans* embryo, and that PAB-1 and PAB-2 are essential for both poly tail-dependent and -independent silencing aspects of the mechanism. T_{1/2} correspond to the half deadenylation time. To test this, deadenylation assays were performed on RL-6-mi R-35-p A86 reporters bearing a shorter 32nt linker, or a longer 262nt linker ahead of the poly tail.

Mentioned biomedical entities:

Genes: AIN-1, AIN-2, ALG-1, ALG-2, Argonaute, CNOT1, GST, HD2, IFG-1, L32, PAB-1, PAB1, PABP, PABPC1, PAIP2, Pabpc1, S1, TBB-2, ain-1, cog-1, let-7, lin-41, lsy-6, miR-35, miR-52, pab-2

Proteins: N/A

Diseases: N/A

Chemicals: N/A

Cells: ASE cell, ASEL cell, B, C, N2 Bristol, Plasmid, S1 Worm, cell, cellular, embryonic cell-free, embryonic lysates, fly embryonic extract, miR-35-specific, stem

Organisms: *C. elegans*, *C. elegans* PAB-1, *C. elegans* PABP, *C. elegans* embryonic, Human, RNA columns, human cytoplasmic PABPC1, mice, mouse PABP, rabbits, zebrafish, zebrafish PABP, zebrafish embryos

Tissues: layer

Pathways: N/A